

AI-Based Paddy Seed Quality Assessment And Farmer Feedback System

*A thesis submitted in partial fulfilment of the requirements
for the award of the degree of*

Bachelor of Technology

By

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Under the Guidance of

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Assistant Professor



**Department of
Computer Science & Engineering**

Manipur Technical University

June, 2026

DEDICATION

This thesis is dedicated to all those who have inspired, supported and stood by us throughout the journey of this project.

To our families: Your unwavering love, patience, and encouragement have been the bedrock of our emotional strength during the most challenging phases. *Your belief in our potential has fortified our resilience and determination.*

To our mentors and instructors: Your expertise, thoughtful guidance, and commitment to our academic growth have been instrumental in shaping both this project and our understanding of the field. *Your encouragement and constructive feedback have been invaluable at each stage.*

To our friends and peers: Your camaraderie, moral support, and motivation have illuminated our path. *Your presence transformed moments of stress into opportunities for shared growth and learning.*

To the global community of researchers, engineers, and innovators in Artificial Intelligence and Computer Vision: Your pioneering work has laid the foundation for this thesis. *Your dedication to advancing knowledge continues to inspire and propel us to contribute meaningfully to this dynamic field.*

This thesis stands as a testament to the collective support, shared vision, and unwavering encouragement of all who have been part of our academic and personal journey.

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MANIPUR TECHNICAL UNIVERSITY

(A University established under the

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Imphal, Manipur – 795004

BONAFIDE CERTIFICATE

This is to certify that the thesis entitled “**AI-Based Paddy Seed Quality Assessment and Farmer Feedback System**” submitted by:

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to the **Department of Computer Science & Engineering**, in partial fulfilment of the requirements for the award of the **Bachelor of Technology** degree in **Computer Science and Engineering**, is a genuine and original work carried out by them under my supervision during the academic year **2025–2026**.

The project embodies the results of their own research and development work and has not been submitted, either in part or in full, for the award of any other degree or diploma of this university or any other institution. The work has been carried out with sincerity, academic integrity, and adherence to research ethics.

We consider this work worthy of consideration for the partial fulfillment of the requirements for the said degree.

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Head of Department

Ma'am Tayenjam Aarena
Assistant Professor & Head,
Computer Science & Engineering

DECLARATION

I hereby declare that the thesis entitled “**AI-Based Paddy Seed Quality Assessment And Farmer Feedback System**”, submitted to the **Department of Computer Science and Engineering, Manipur Technical University**, in partial fulfilment of the requirements for the award of the **Bachelor of Technology** degree, is a result of my own work and effort.

I affirm that this work has been carried out with academic honesty and integrity. All sources of information have been properly cited and acknowledged. This thesis has not been submitted earlier, either in part or in full, for the award of any degree or diploma at this or any other institution.

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Date:

Place:

Sincerely,

Joymangol

Chingangbam

Nilambar Elangbam

Dayananda Laishram

Khagembam Yaiphaba

ABSTRACT

AI-Based Paddy Seed Quality Assessment and Farmer Feedback System is a web platform that addresses counterfeit and substandard paddy seeds which is a recurring cause of crop failure and income loss for farmers across northeastern states of India. Farmers can upload or photograph seed samples through the interface; the system analyzes visual characteristics using CNN-based models running on cloud inference and returns a quality summary before sowing decisions are made, with no lab equipment required.

The platform also includes a Farmer Feedback and Complaint Module, where users can report poor germination, suspected counterfeits, or crop failures. Officers access an analytics dashboard that aggregates these reports to surface risk patterns at the batch and regional level.

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Symbols and Acronyms

$\approx$	Approximately equal to, used in performance evaluation
$\leq$	Less than or equal to, used in joint angle thresholding
π	Pi constant used in degree-radian conversion
θ	Angle between three landmarks
x, y	Coordinates of a detected keypoint in the frame
2D	2-Dimensional
3D	3-Dimensional
%	Percentage
AI	Artificial Intelligence
BGR	Blue-Green-Red color format
CNN	Convolutional Neural Network
COVID	CoronaVirus Disease of 2019
CV	Computer Vision
FPS	Frames Per Second
GB	Gigabyte
GHUM & GHUML	Generative 3D Human Shape and Articulated Pose Models
GUI	Graphical User Interface
iOS	iPhone Operating System
ML	Machine Learning
OpenCV	Open Source Computer Vision Library

RAM Random Access Memory

RGB Red-Green-Blue color format

Chapter 1

Introduction

1.1 Background and Context

Agriculture remains the foundational backbone of many global economies, heavily influencing food security, employment, and national income. At the core of all agricultural productivity is the quality of the seed planted; it is the most critical and non-negotiable input that determines crop yield, resilience to disease, and overall harvest quality. However, the agricultural sector is currently facing a severe crisis due to the proliferation of counterfeit or "fake" seeds in the market.

Counterfeit seeds are often packaged to resemble premium brands or are visually indistinguishable from high-yield varieties, yet they exhibit drastically lower germination rates and poor genetic traits. Traditional methods of verifying seed authenticity rely on destructive laboratory testing or subjective physical inspection, both of which are inaccessible, expensive, and far too slow for the average smallholder farmer. Consequently, farmers often discover they have been deceived only after planting, leading to catastrophic losses in time, resources, and income.

1.2 Motivation

The primary motivation for this project stems from the devastating socio-economic impact that fake seeds have on farming communities. When farmers unknowingly purchase and plant counterfeit seeds, they face severe financial distress, which cascades into broader issues of local food insecurity and loss of trust in agricultural supply chains. There is an urgent, unmet need for accessible technological tools that empower grassroots agricultural workers to protect themselves against fraud.

Simultaneously, the rapid advancement of deep learning, computer vision, and cloud computing presents an unprecedented opportunity to address this crisis. By leveraging Convolutional Neural Networks (CNNs), it is now possible to achieve laboratory-grade visual analysis automatically. The motivation is to harness these cutting-edge AI technologies and package them into a simple, web-based tool, putting the power of automated, real-time seed verification directly into the hands of the farmers.

1.3 Problem Statement

Despite the devastating effects of agricultural fraud, farmers currently lack a reliable, accessible, and rapid method to accurately distinguish between genuine and counterfeit seeds before the planting season begins. Visual similarities make manual detection nearly impossible, and existing institutional supply chains lack a transparent, real-time verification system.

Therefore, the core problem is the absence of an integrated, technology-driven solution that can automatically analyze seed imagery for subtle fraudulent anomalies, while also capturing localized, crowdsourced feedback from farmers to map and identify the spread of counterfeit seeds in real-time.

1.4 Objectives

The overarching goal of this project is to create an intelligent ecosystem for seed verification. To achieve this, the project outlines the following specific objectives:

1. To design and train a robust deep learning model (utilizing architectures such as ResNet) capable of accurately classifying seed images and identifying visual anomalies associated with counterfeit seeds.
2. To develop and deploy a highly accessible, user-friendly web-based platform that allows farmers to seamlessly upload images of their seeds for real-time AI inference and verification.
3. To integrate a comprehensive "Farmer Feedback System" within the platform,

enabling users to report their post-purchase and post-planting experiences (such as germination failure rates).

4. To utilize the crowdsourced feedback and newly uploaded images to continuously refine the database, creating an early warning system against emerging counterfeit supply chains.

1.5 Scope of the Project

The scope of this project encompasses the end-to-end development of an AI-driven web application tailored for agricultural quality control. This includes the collection, curation, and preprocessing of a targeted seed image dataset, followed by the training, validation, and optimization of a Convolutional Neural Network (CNN).

Additionally, the scope covers the full-stack development of the web platform, including an intuitive frontend interface for image uploads and a robust backend to handle model inference and database management. The project is strictly limited to the non-destructive, visual assessment of seeds via digital imagery; it does not extend to biochemical, genetic, or DNA-based testing methods. The final deliverable will be a functional prototype that demonstrates the feasibility of combining deep learning with crowdsourced feedback to combat agricultural fraud.

Chapter 2

Literature Review

2.1 Seed Quality Assessment: Traditional Methods

Conventional seed quality assessment relies on two broad approaches: manual visual inspection and laboratory-based biochemical testing. In field practice, inspectors evaluate seeds by examining size, shape, colour, and surface texture. These judgements are inherently subjective, vary between observers, and are difficult to standardise across districts or seasons. Laboratory tests such as the Tetrazolium (TZ) test and standard germination assays provide more defensible viability estimates, but both are destructive, the seed cannot be planted after testing and typically require four to seven days to complete under controlled conditions. This turnaround makes them impractical for screening large consignments at intake points, or for use by farmers who need a decision before the planting window closes. Near-infrared (NIR) spectroscopy and X-ray transmission imaging have been investigated as non-destructive alternatives. NIR detects internal moisture gradients and biochemical composition without damaging the sample; X-ray imaging can reveal embryo defects invisible on the seed surface. Both methods perform well in laboratory conditions, but the equipment costs typically **USD 15,000–80,000** per unit and the need for trained operators confine their use to national seed-testing stations. Smallholder farmers and district-level inspectors in regions such as north-east India have no practical access to either technology, which is precisely where the need for affordable, rapid screening is most acute.

2.2 Deep Learning for Agricultural Image Classification

Convolutional Neural Networks (CNNs) have been applied to a range of agricultural

classification problems over the past decade, including leaf disease identification [1], weed recognition [2], and post-harvest quality grading [3]. Their main advantage over classical machine-learning pipelines is that spatial features are learned directly from raw pixel data during training, removing the need to manually engineer feature extractors for each new crop or defect type.

Applying CNNs to seed classification follows naturally from this body of work. Paddy seeds, for example, exhibit subtle differences in surface texture and colour between certified and counterfeit lots differences that are difficult to quantify with handcrafted descriptors but emerge reliably from sufficiently large training sets. The computational cost of running a trained CNN at inference time is also modest enough to support server-side web deployment, which matters when the target users have intermittent electricity but reasonable mobile-data coverage.

2.3 Transfer Learning and ResNet Architectures

Training a CNN from random initialization requires on the order of tens of thousands of labeled examples per class to achieve stable convergence. Annotated seed image datasets of that scale do not yet exist for most crop varieties, and certainly not for the specific task of detecting counterfeits. Transfer learning is the standard response: a model pre-trained on ImageNet (1.28 million labeled images across 1,000 categories) is fine-tuned on the smaller domain-specific dataset. The early convolutional layers, which encode low-level features such as edges and textures, are retained; only the later task-specific layers are updated. In practice, this approach consistently reaches accuracy close to what a from-scratch model would achieve on a dataset roughly ten times larger.

ResNet architectures are well-suited to this fine-tuning scenario. The residual skip connections introduced by He et al. [4] allow gradients to propagate through very deep networks without vanishing, making it feasible to fine-tune models with 50 or more layers on datasets that would cause shallower architectures to overfit. ResNet-50 has become a common baseline in agricultural image classification because it balances parameter count (≈ 25 million) against accuracy on standard benchmarks, and pre-trained weights are publicly available. For this project, it provides a well-validated starting point that reduces both training time and the risk of overfitting on the available paddy seed dataset.

2.4 Farmer Feedback Systems

A detection model trained on a fixed dataset will degrade over time if counterfeiting practices evolve and no new training data are collected. Field-level feedback from farmers is one practical mechanism for capturing these shifts. When a farmer submits a seed batch for image-based verification and later reports germination failure or abnormal yield from that batch, the report constitutes a labeled example a model prediction paired with a ground-truth outcome that can be incorporated into periodic retraining cycles.

Feedback also serves a more immediate function. Aggregating reports across users can flag a suspect lot or supplier before the model itself is updated: three independent failure reports from the same district within a fortnight is a meaningful signal, regardless of what the image classifier predicts. This kind of rule-based alert complements the model rather than replacing it. The practical difficulty is sustaining farmer participation; most deployments in low-resource contexts find that single-tap rating interfaces produce far higher response rates than free-text fields or structured forms [5].

2.5 Web-Based Agricultural Platforms

Deploying a trained model as a web service separates the computational workload from the end user’s device. A farmer uploads a seed image through a browser; the server runs inference and returns a result. This architecture has two concrete advantages over distributing a standalone mobile application: the model can be updated centrally without requiring client-side installations, and every prediction is logged in a structured format that supports monitoring and retraining.

For farmers in Manipur and similar regions, browser-based access via a mid-range Android handset is more realistic than assuming a dedicated app installation. Keeping the frontend lightweight and avoiding heavy JavaScript bundles and large media assets reduces page-load times on 4G connections with variable signal quality. Containerised (Dockerized) inference endpoints can handle the computationally intensive tasks without requiring a permanently running server, which reduces operating costs during off-season periods when query volume is low. Together, these design choices bring ML-assisted quality screening within reach of users who would otherwise have no access to any testing infrastructure.

2.6 Review of Related Work

Early automated seed classification used Support Vector Machines (SVMs) and Random Forests operating on handcrafted features derived from shape measurements, colour histograms, and texture descriptors. Ni et al. [6] classified rice seed varieties using morphological parameters extracted from binary silhouettes, achieving 94% accuracy across five varieties under controlled lighting. These methods perform adequately when intra-class variation is low and imaging conditions are uniform, but generalise poorly to photographs taken in field settings.

Deep learning approaches have largely replaced handcrafted methods in more recent work. Liu et al. [7] applied VGG16 transfer learning to maize seed defect detection and reported 97.3% accuracy on a 12-class dataset. Xu et al. [8] used ResNet-50 for soybean quality grading and found that fine-tuning from ImageNet weights outperformed training from scratch by approximately 8 percentage points when training data were limited to fewer than 2,000 images per class. However, both studies address naturally occurring defects — cracking, mould, insect damage rather than intentional substitution. A counterfeit seed is designed to resemble the genuine article; distinguishing the two is a harder classification problem than separating a damaged kernel from an intact one, because the visual gap is smaller and the adversarial nature of counterfeiting means it may shift over time. No published study directly evaluates CNN performance on verified counterfeit-versus-genuine seed pairs, and none integrates a structured feedback collection mechanism into the evaluation pipeline.

2.7 Summary and Research Gaps

The reviewed literature establishes that transfer-learned CNNs are effective for agricultural image classification, including seed quality assessment. Three specific gaps are directly relevant to this work. First, no published study has targeted intentional counterfeiting as distinct from natural quality defects; the two problems differ in that counterfeits may pass casual visual inspection and require the model to detect very subtle cues. Second, existing models are evaluated on laboratory-quality images collected under controlled conditions; performance on photographs taken by farmers using mobile devices in variable lighting has not been reported. Third, no end-to-end system has been published that connects model inference to a structured feedback collection mechanism and uses that feedback to retrain the model iteratively.

This project addresses all three gaps. The classifier is trained and evaluated specifically on a fake-versus-genuine paddy seed dataset; the web platform is tested

with mobile-captured images; and the farmer feedback module is built directly into the application, with logged responses stored in a format suitable for future retraining rounds.

Chapter 3

System Design and Architecture

3.1 Overall System Architecture

The AgriVerify platform is engineered as a distributed, multi-tier web system that integrates deep learning inference, cloud-based AI orchestration, and a role-aware web application into a single coherent service. The architecture adheres to the **Clean Architecture** principle (Onion Architecture variant), which ensures that the core business logic remains entirely independent of infrastructure concerns such as databases, cloud vendors, and UI frameworks. Dependencies flow strictly inward: the outer layers (Infrastructure and API) depend on abstractions exposed by the inner layers (Application and Domain), never the reverse [9].

At the highest level, the platform comprises three major subsystems:

1. **Deep Learning Microservice** — A PyTorch/ResNet-50 model trained on a 1,563-image paddy seed corpus and served as a containerised FastAPI endpoint on Google Cloud Platform (GCP) Cloud Run [4].
2. **Backend API** — An ASP.NET Core 8 Web API acting as the primary orchestration layer. It authenticates users, mediates access to the ML microservice and Azure AI services (Custom Vision, AI Vision OCR, and Azure OpenAI), persists results to Microsoft SQL Server via Entity Framework Core, and exposes a versioned REST API consumed by the frontend [10].
3. **Frontend Web Application** — A Next.js 14 App Router application providing role-based dashboards for Farmers, Officers, and Administrators. All browser-to-backend traffic is forwarded through a server-side proxy route to centralise authentication token injection and prevent credential exposure in the browser [11].

3.1.1 Backend Architectural Layers

The backend follows the four-layer Onion model, each layer encapsulating a distinct concern:

Domain Layer Contains core entities (`User`, `VerificationHistory`, `ProductComplaint`), value objects, enumerations (`UserRole`, `VerificationStatus`, `ComplaintStatus`), and domain exceptions. This layer carries no dependency on any external framework or service.

Application Layer Orchestrates business workflows through application services (`VerificationService`, `UserService`, `ComplaintService`), defines Data Transfer Objects (DTOs), AutoMapper profiles, and FluentValidation rules. It depends only on the Domain layer.

Infrastructure Layer Provides concrete implementations for interfaces declared in the Application layer: Entity Framework Core repositories, the `ApplicationDbContext`, Azure service wrappers (`CustomVisionService`, `ComputerVisionService`, `OpenAIService`, `BlobStorageService`), and migration management. This layer bridges the application to external systems without surfacing implementation details to upper layers.

API Layer Hosts ASP.NET Core controllers (`AuthController`, `VerificationController`, `ComplaintController`, `AdminController`), the JWT middleware pipeline, Swagger/OpenAPI documentation, and the global exception handling middleware.

3.2 Functional Requirements

The functional requirements presented below were derived from stakeholder analysis involving farmers, agricultural officers, and domain experts. They form the contractual basis for all implementation decisions.

3.2.1 FR-01: User Registration and Authentication

- Farmers must register using their name, email address, state, and district, and authenticate via an email–password credential pair.
- Agricultural officers must authenticate using an officer ID and a time-based one-time password (TOTP) as a second authentication factor.

- Administrator accounts are provisioned through database seeding; credentials are managed by the platform.
- The system must issue JWT access tokens and refresh tokens upon successful login, and must invalidate both on logout.

3.2.2 FR-02: Seed Verification

- Authenticated farmers must be able to upload front and back images of seed packaging for AI-powered authenticity analysis.
- The system must perform parallel inference using Azure Custom Vision for image classification, Azure AI Vision for OCR text extraction, and Azure OpenAI for contextual report synthesis.
- A confidence score, a verification status (Genuine / Counterfeit / Suspicious), extracted OCR text, and a natural-language summary must be returned to the caller and persisted to the database.
- An anonymous public verification endpoint must be available to unauthenticated users.

3.2.3 FR-03: Complaint Management

- Authenticated farmers must be able to file complaints against seed batches, specifying issue type, affected district, severity level, and optional image evidence.
- Officers must be able to view assigned complaints, update their status (Open / Under Investigation / Resolved / Closed), and record investigation actions with timestamps.
- Administrators must be able to view all complaints, filter by district and status, and assign unassigned complaints to available officers.

3.2.4 FR-04: Role-Based Dashboards

- Farmers must have access to a personal dashboard displaying their verification history, complaint status, and summary metrics.
- Officers must be able to view a complaint queue, perform batch risk assessments, and access officer-level analytics.

- Administrators must have access to platform-wide analytics, user management controls, and reference data configuration.

3.2.5 FR-05: Batch Risk Registry

- The system must maintain a `BatchRiskRegistry` that aggregates complaint statistics per batch number, tracks affected districts, computes average severity scores, and assigns categorical risk levels.

3.2.6 FR-06: Deep Learning Model Inference

- The ResNet-50 microservice must accept seed images, apply the 224×224 preprocessing pipeline, perform forward inference, and return class probabilities alongside 17 extracted DUS (Distinctness, Uniformity, and Stability) physical parameters such as length, width, colour, texture, and shape metrics.
- The microservice must perform variety matching against a reference database of 12 registered paddy varieties and return the top-3 matches with their similarity percentages.

3.3 Non-Functional Requirements

Non-functional requirements govern the quality attributes of the system and are treated as first-class design constraints throughout the implementation.

NFR-01 — Performance The end-to-end verification pipeline (image upload $\rightarrow$ AI orchestration $\rightarrow$ database write $\rightarrow$ response) must complete within a latency budget suitable for interactive web use. Parallel execution of OCR and Custom Vision calls via `Task.WhenAll` is mandated to reduce blocking latency. The FastAPI microservice must serve single-image inference in sub-second wall-clock time on GCP Cloud Run hardware.

NFR-02 — Scalability The backend must be stateless and deployable as a Dockerised container, enabling horizontal scaling without code changes [12]. The Next.js frontend must support server-side rendering and edge caching.

NFR-03 — Security All HTTP traffic must be served over TLS (HTTPS). JWT access tokens must carry minimal claims and expire within a configurable time-to-live (TTL). Passwords must be stored as cryptographic hashes using

PBKDF2 or bcrypt. Sensitive credentials (Azure keys, database connection strings, and API keys) must reside in environment variables and must never be hard-coded. Role-based endpoint protection must be enforced through ASP.NET Core `[Authorize]` policies [13].

NFR-04 — Reliability and Data Integrity All grouped repository operations must execute within a Unit of Work transaction with atomic commit-or-rollback semantics. Soft-delete functionality must be implemented for complaint and verification records to support future legal auditing. Health check endpoints (`/health` and `/api/health`) must expose the connectivity status of all critical dependencies.

NFR-05 — Maintainability The codebase must adhere to Clean Architecture layering with clear separation of concerns. Infrastructure services must be registered through an `InfrastructureServiceCollectionExtensions` extension class rather than inline in `Program.cs`. All public API endpoints must be documented via Swagger/OpenAPI 3.0.

NFR-06 — Usability and Accessibility The frontend must deliver a responsive layout usable on both desktop browsers and mobile devices, supporting farmers who access the platform via smartphones. The UI must provide immediate loading feedback through skeleton placeholders and toast notifications for asynchronous operations.

NFR-07 — Portability The deep learning microservice must be exported in multiple formats (`.pth`, TorchScript `.pt`, and ONNX `.onnx`) to support deployment across heterogeneous runtime environments. The entire application stack must be containerisable via Docker.

3.4 Tools and Technologies

3.4.1 Deep Learning Stack (PyTorch and ResNet-50)

The classification model is implemented entirely in **Python 3.10** using the following libraries [14]:

- **PyTorch** (`torch`, `torchvision`) — The primary deep learning framework used for model definition, training, and inference. PyTorch’s dynamic computation graph simplifies debugging and permits flexible architecture modifications during transfer-learning fine-tuning phases.

- **ResNet-50** — A 50-layer deep residual network pre-trained on ImageNet (1.2M images, 1,000 classes) sourced from `torchvision.models` [4]. The backbone contributes approximately 23.5M parameters; a custom three-layer classification head ($2,048 \rightarrow 512 \rightarrow 128 \rightarrow 3$ units) adds a further 1.1M trainable parameters.
- **OpenCV** (`cv2`) — Used in the DUS parameter extraction pipeline for contour detection, pixel-to-millimetre calibration, and morphological analysis of seed images to compute the 17 physical parameters (colour mean/standard deviation, circularity, solidity, grain length/width, and related metrics) [15].
- **NumPy** / **Pandas** — Employed for numerical array manipulation and dataset statistics management [16].
- **scikit-learn** — Provides stratified train/validation/test split utilities and evaluation metric helpers [17].
- **Matplotlib** / **Seaborn** — Used for training curve visualisation and confusion matrix rendering.

Training was conducted on a GPU-enabled environment using the **AdamW** optimiser with weight decay $\lambda = 10^{-4}$, cross-entropy loss with label smoothing $\varepsilon = 0.1$, and a two-phase transfer learning schedule. Phase 1 (Epochs 1–8, backbone frozen, learning rate $= 10^{-3}$) allows the classification head to converge quickly, while Phase 2 (Epochs 9–15, full fine-tuning, learning rate $= 10^{-4}$) enables end-to-end weight adaptation. The final ResNet-50 model achieved a test accuracy of **99.57%** on the combined 1,563-image dataset, with a train-to-validation accuracy gap below 5%.

3.4.2 Backend Stack (ASP.NET Core 8, SQL Server, and Azure)

The backend REST API is built on the **Microsoft .NET 8** ecosystem. Table 3.1 summarises the key technology components and their roles within the system.

Azure AI services are consumed through dedicated service wrappers (`CustomVisionService`, `ComputerVisionService`, and `OpenAIService`) registered in the dependency injection container as singletons. The parallel orchestration pattern (`await Task.WhenAll(ocrTask, classificationTask)`) ensures that OCR extraction and image classification execute concurrently, minimising per-request latency [10].

Table 3.1: Backend Technology Stack

Component	Technology / Library
Web Framework	ASP.NET Core 8.0 / .NET 8 (Kestrel)
Database	Microsoft SQL Server
ORM	Entity Framework Core (Code-First, Migrations)
Authentication	ASP.NET Core Identity + JWT Bearer
Object Mapping	AutoMapper
Validation	FluentValidation + DataAnnotations
Logging	Serilog (Console and File sinks)
API Documentation	Swagger / OpenAPI 3.0
AI Services	Azure Custom Vision, Azure AI Vision (OCR), Azure OpenAI
Cloud Storage	Azure Blob Storage
Deployment	Docker / Render Cloud

3.4.3 Model Serving (FastAPI and Docker)

The trained PyTorch model is wrapped in a **FastAPI** microservice and deployed as a Docker container on **GCP Cloud Run** [18]. The inference pipeline within the microservice follows five sequential stages:

1. **Request** — The FastAPI endpoint receives a multipart image payload from the ASP.NET backend.
2. **Preprocessing** — The image is resized to 224×224 pixels and normalised using the ImageNet channel-wise mean $\mu = [0.485, 0.456, 0.406]$ and standard deviation $\sigma = [0.229, 0.224, 0.225]$.
3. **Model Inference** — The ResNet-50 model processes the normalised tensor and produces class logits.
4. **Post-processing** — Softmax converts logits to class probabilities; OpenCV routines extract the 17 DUS parameters; variety matching is performed against the 12-variety reference database.
5. **Response** — A structured JSON payload containing the prediction label, confidence score, physical parameters, DUS classification, and top-3 variety matches is returned to the backend.

The model is exported in three formats to maximise deployment flexibility: PyTorch native format (`.pth`, ≈ 98 MB), TorchScript (`.pt`, ≈ 97 MB, for C++ runtimes),

and ONNX (.onnx, ≈ 98 MB, for cross-platform inference). Continuous delivery is automated via a **GitHub Actions** CI/CD pipeline that triggers a Docker build-and-push to GCP Artifact Registry on every commit to the main branch.

3.4.4 Frontend Stack (Next.js, TypeScript, and TanStack Query)

The user-facing application is a **Next.js 14 App Router** single-page application written in **TypeScript**. The key technology choices are described below.

- **Next.js 14 App Router** — Provides file-system-based routing with route groups that cleanly separate the three role namespaces (`(farmer)`, `(officer)`, and `(admin)`). Server components handle initial data fetching; client components manage interactive UI behaviour.
- **TypeScript** — Enforces strict compile-time type safety across all layers, including shared DTO interfaces (`src/types/`), domain service modules (`src/services/`), and custom React hooks (`src/hooks/`).
- **TanStack Query (React Query v5)** — Manages all asynchronous server state, including cache invalidation, background refetch, pagination, and optimistic mutation rollback. Query hooks are co-located with domain modules; the global `QueryClient` disables refetch-on-window-focus to reduce incidental network requests [19].
- **Zustand** — A lightweight client-side state store for authentication state (`user`, `token`, `role`, `isAuthenticated`). The store is initialised from `localStorage` on the client side and exposes `setAuth` and `clearAuth` actions [20].
- **Axios** — Centralised HTTP client configured with a base URL of `/api/proxy`. A request interceptor automatically injects the Bearer token from storage; a response interceptor handles 401 errors by transparently attempting a token refresh before replaying the failed request.
- **Next.js API Route Proxy** — A server-side proxy route at `/api/proxy/[...path]` forwards all browser requests to the ASP.NET backend using the URL stored in the `BACKEND_API_URL` environment variable, preventing credential exposure in the browser.
- **Tailwind CSS** — Utility-first styling framework with extended design tokens (colours, border radii, and shadows) defined in `tailwind.config.ts`.

- **Shadcn/UI + Custom Primitives** — A compact local component library providing Button, Input, Select, Dialog, Sheet, Table, Badge, Tabs, Skeleton, and Progress components built on native HTML elements with minimal abstraction overhead.

3.5 System Architecture Diagrams

3.5.1 High-Level Architecture

Figure 3.1 illustrates the end-to-end system topology. The user’s browser communicates exclusively with the Next.js frontend server; no backend URLs are ever exposed to the client. All backend calls are proxied server-side to the ASP.NET Core API, which delegates to the ML microservice on GCP and to the suite of Azure AI services. Verification records are persisted in Microsoft SQL Server via Entity Framework Core, and seed images are stored in Azure Blob Storage.

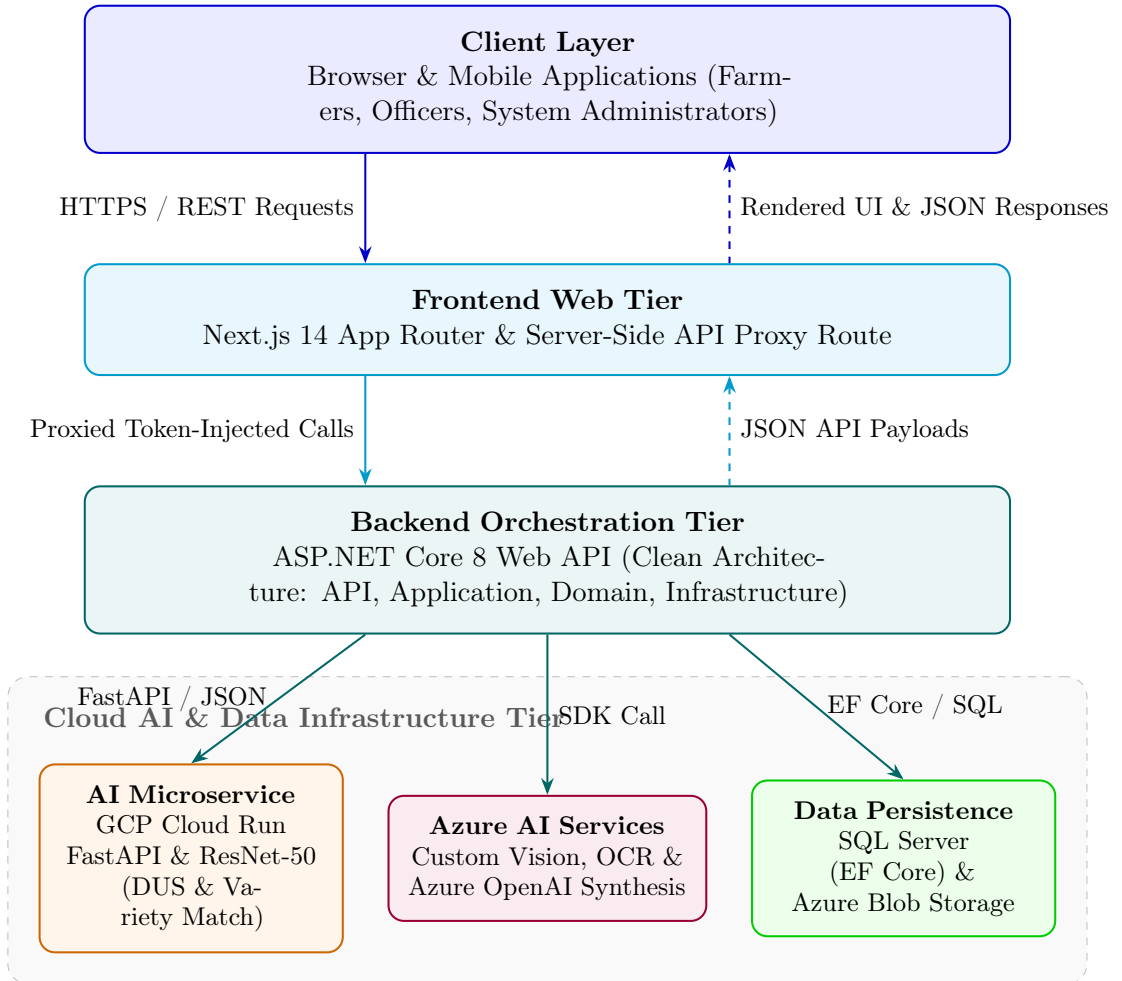


Figure 3.1: High-level system architecture of the AgriVerify platform. Solid arrows indicate request direction; dashed arrows indicate response direction.

3.5.2 Data Flow Diagram

Figure 3.2 traces the data flow for the primary seed verification use case. When a farmer submits seed images, the request travels from the browser through the Next.js proxy, enters the ASP.NET verification pipeline, and is fanned out in parallel to the ML microservice and two Azure AI services. The combined results are synthesised by Azure OpenAI, persisted to the database, and returned as a structured JSON response.

3.5.3 Entity-Relationship Diagram

Figure 3.3 presents the simplified entity-relationship (ER) diagram for the core domain model persisted in Microsoft SQL Server. The model captures the principal entities and the cardinality of their associations.

Several design decisions embedded in the ER model merit explicit discussion:

- **Soft delete** — `IsDeleted` and `DeletedAt` fields on `VerificationHistory` and `ProductComplaint` preserve audit trails and support future legal review without physically removing records.
- **Third Normal Form (3NF) normalisation** — Seed package information, verification records, and complaint data are stored in separate tables, eliminating data redundancy and update anomalies.
- **Indexed columns** — `User.Email`, `VerificationHistory.UserId`, `ProductComplaint.Status`, and `ProductComplaint.District` carry non-clustered indices to support efficient dashboard query execution.
- **BatchRiskRegistry** — Updated by a background aggregation job triggered on every complaint state change, this denormalised table provides fast, pre-computed risk summaries for officer-facing analytics without requiring expensive real-time aggregation queries.

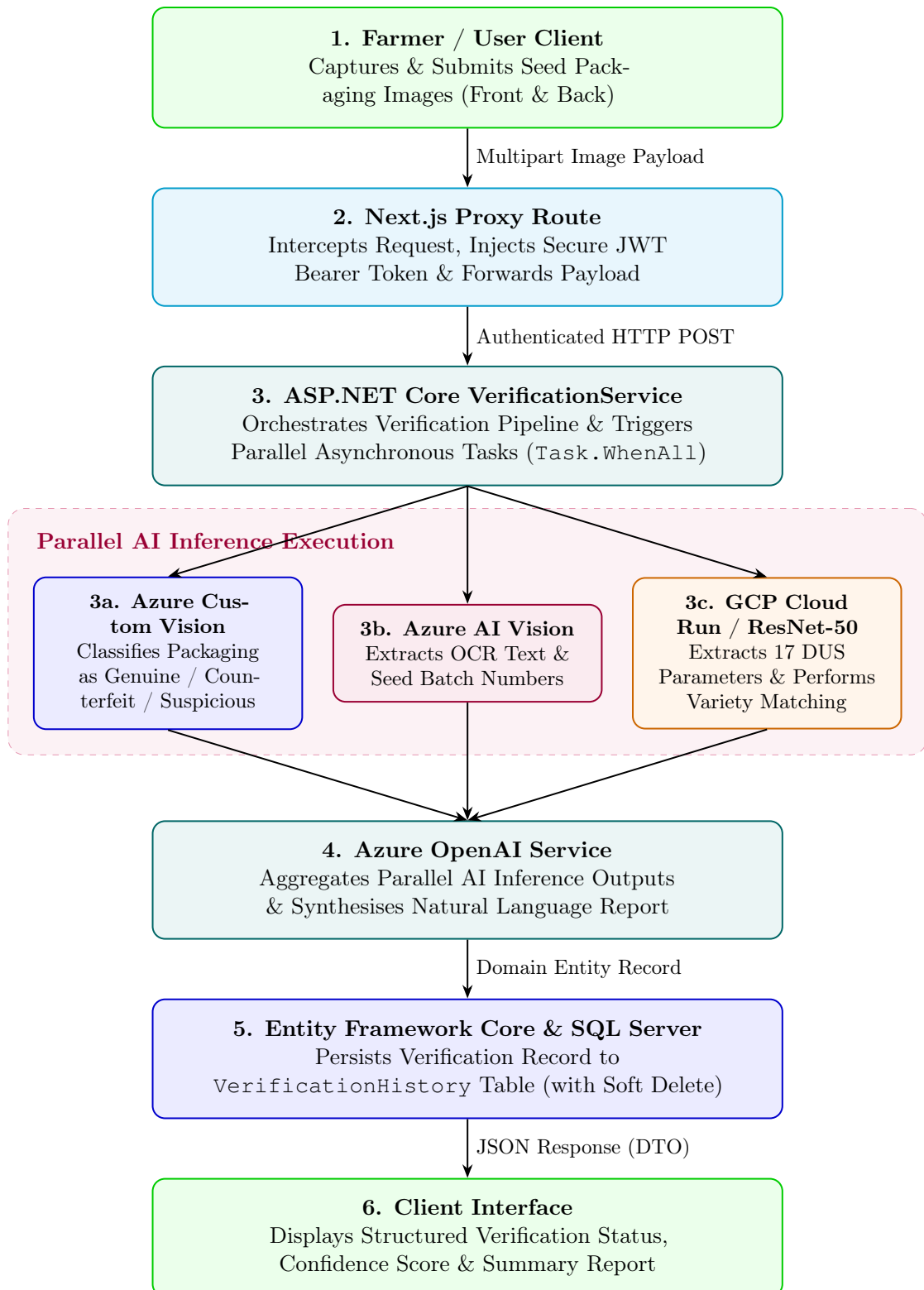


Figure 3.2: Data flow diagram for the seed verification pipeline. The three inference calls (Custom Vision, OCR, ResNet-50) execute in parallel; results are aggregated before OpenAI synthesis.

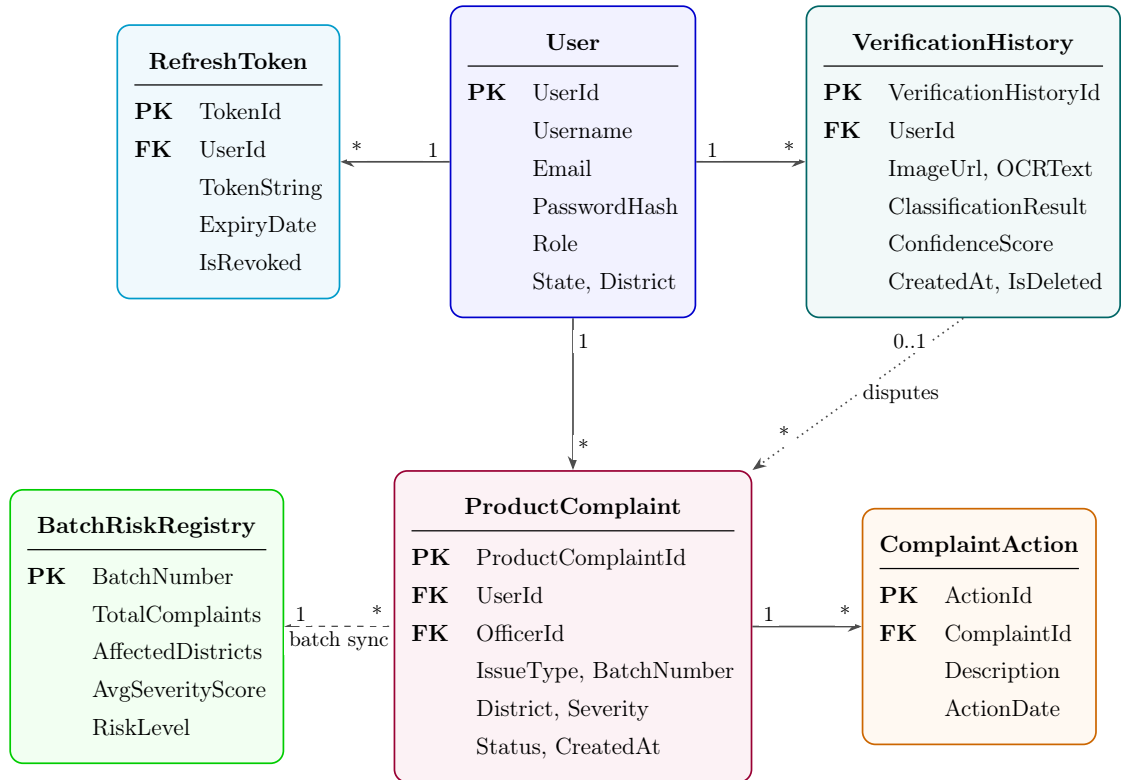


Figure 3.3: Simplified entity-relationship diagram for the AgriVerify domain model. Cardinality notation follows UML conventions.

Chapter 4

Deep Learning Model for Paddy Seed Quality Assessment

4.1 Dataset Description

4.1.1 Data Sources and Collection (Kaggle)

The dataset used in this project was assembled from two distinct sources on Kaggle: one publicly available and one custom-collected by our team in collaboration with the Indian Council of Agricultural Research (ICAR), Manipur. Combining these two sources improved both the diversity and the real-world representativeness of the training data, ensuring that the model is exposed to paddy seed images captured under a variety of conditions and geographic contexts.

4.1.1.1 Public Source

The first source is the Paddy Seeds Quality Classification dataset published on Kaggle by Ayyan Arkadalkani:

```
https://www.kaggle.com/datasets/ayyanarkadalkani/  
paddy-seeds-quality-classification-dataset
```

This dataset comprises 1,213 labeled images distributed across three quality classes—Healthy, Damaged, and Fake/Low Quality—with a total storage footprint of approximately 183 MB. It provided a strong baseline of labeled examples captured under varied lighting and background conditions, making it suitable as a general-purpose foundation for transfer learning.

4.1.1.2 Custom Source

The second source is a custom dataset collected directly from ICAR, Manipur. Images of paddy seeds were captured under controlled laboratory conditions at the ICAR facility to reflect the specific seed varieties cultivated in the northeastern region of India. This collection totals 350 images and occupies approximately 24.7 MB of storage. The dataset has been published on Kaggle as an open-access resource to support future research in regional paddy seed analysis:

[https://www.kaggle.com/datasets/nilambarelangbam/
paddy-seed-images-collected-from-icar-manipur](https://www.kaggle.com/datasets/nilambarelangbam/paddy-seed-images-collected-from-icar-manipur)

By merging both datasets, we obtained a combined corpus of 1,563 images that is both larger and more geographically diverse than either source alone, better reflecting the range of seed conditions encountered in real agricultural settings across India.

4.1.2 Dataset Statistics and Class Distribution

After merging both datasets, the combined corpus contains 1,563 paddy seed images distributed across three quality classes. Table 4.1 summarises the source-wise composition, and Table 4.2 presents the class-wise distribution.

Table 4.1: Dataset Source Summary

Source	Images	Size	Kaggle Identifier
Paddy Seeds Quality Classification (Ayyan Arkadalkani)	1,213	~183 MB	ayyanarkadalkani/ paddy-seeds-quality-...
ICAR Manipur Custom Dataset (Our Collection)	350	~23.7 MB	nilambarelangbam/ paddy-seed-images-...
Combined Total	1,563	~208 MB	—

Table 4.2: Class-wise Distribution of the Combined Dataset

Class	Number of Images	Percentage (%)
Healthy Paddy Seeds	~750	~48.0
Damaged Paddy Seeds	~510	~32.6
Fake/Low Quality Seeds	~303	~19.4
Total	1,563	100.0

The dataset exhibits a moderate class imbalance, with Healthy seeds being the most represented class and Fake/Low Quality seeds the least. To prevent the model from developing a bias toward the majority class, weighted random sampling was applied during training, ensuring that each class contributes proportionally to every training batch regardless of its raw frequency in the dataset.

4.1.3 Train/Validation/Test Split Strategy

We partitioned the 1,563 images into three non-overlapping subsets following a stratified split strategy. Stratified splitting ensures that the class proportions observed in the full dataset are preserved in each subset, preventing any split from being inadvertently dominated by a single quality class. Table 4.3 summarises the split.

Table 4.3: Dataset Split Statistics

Split	Proportion	Approx. Images	Purpose
Training Set	80%	~1,250	Model weight optimisation via back-propagation
Validation Set	10%	~157	Hyperparameter tuning and early stopping
Test Set	10%	~156	Final unbiased evaluation on unseen data

The training set is used exclusively for updating model weights. The validation set is monitored at the end of every epoch to guide hyperparameter decisions and to detect the onset of overfitting. The test set is held out entirely until the final evaluation stage, ensuring that all reported performance metrics reflect genuine generalisation to previously unseen examples.

4.1.4 Data Preprocessing and Augmentation

Before feeding images into the network, a standard preprocessing pipeline was applied to ensure compatibility with the ResNet50 architecture. All images were rescaled to 224×224 pixels—the standard input resolution for ImageNet-pretrained ResNet models. Pixel values were then normalised using the ImageNet channel-wise mean and standard deviation:

$$\mu = [0.485, 0.456, 0.406], \quad \sigma = [0.229, 0.224, 0.225] \quad (4.1)$$

to align the input distribution with that seen during backbone pre-training.

To expand the effective diversity of the training set and improve the model’s robustness to real-world image variability, eight data augmentation techniques were

applied exclusively during training:

1. **Random Resized Crop**—Randomly crops a region of the image and resizes it to 224×224 pixels, simulating varying camera distances and seed positions within the capture frame.
2. **Horizontal Flip**—Randomly mirrors the image along the vertical axis with probability 0.5, exploiting the bilateral symmetry common to seed grains.
3. **Vertical Flip**—Randomly mirrors the image along the horizontal axis with probability 0.5, further increasing orientation diversity in the training set.
4. **Random Rotation**—Rotates the image by a randomly sampled angle within $\pm 30^\circ$, accounting for seeds that are not perfectly aligned during imaging.
5. **Translation/Random Shift**—Randomly shifts the image content along the x and y axes, preventing the model from relying on the absolute position of the seed within the frame.
6. **Colour Jitter**—Randomly perturbs brightness, contrast, saturation, and hue within specified bounds, enhancing robustness to lighting variations in agricultural settings.

4.2 Model Architecture

4.2.1 ResNet50 Backbone (25.6M Parameters)

ResNet50 was selected as the backbone architecture for this classification task. This deep residual network contains 50 layers and 25.6 million trainable parameters. The architecture is built upon residual blocks, which employ skip connections to enable the training of very deep networks by mitigating the vanishing gradient problem.

The pre-trained weights from the ImageNet dataset were used as the initial checkpoint, providing a rich set of learned feature representations for natural images. Transfer learning was then applied: the backbone layers were initially frozen during Phase 1 training, with only the custom classification head being optimised. This approach leverages the generalisable features learned on ImageNet while adapting the model to the specific task of seed quality classification.

4.2.2 Custom Classification Head

Upon the frozen ResNet50 backbone, a custom classification head was appended to perform the final three-way classification (Healthy, Damaged, Fake/Low Quality). The head comprises:

1. Global Average Pooling layer (reducing the spatial dimensions from 7×7 to a single vector)
2. Fully connected layer with 512 hidden units
3. Batch Normalisation layer
4. ReLU activation function
5. Dropout layer (dropout rate: 0.5)
6. Output layer with 3 units (one per class) and softmax activation

This architecture ensures that the model learns class-specific patterns while maintaining numerical stability during training.

4.2.3 Anti-Overfitting Techniques

To prevent overfitting and improve generalisation to unseen data, multiple regularisation techniques were employed:

- **Dropout**—Applied in the classification head (rate: 0.5) to randomly deactivate neurons during training, reducing co-adaptation.
- **Batch Normalisation**—Applied after linear transformations to normalise layer inputs, stabilising training and acting as a regulariser.
- **Label Smoothing**—Cross-entropy loss was computed with label smoothing ($\epsilon = 0.1$), preventing the model from assigning overconfident predictions.
- **Weight Decay (L2 Regularisation)**—A weight decay coefficient of 0.0001 was applied to all trainable parameters, penalising large weights.
- **Data Augmentation**—As described in Section 4.1.4, eight augmentation techniques were applied to increase training set diversity.
- **Early Stopping**—Training was monitored on the validation set, and the best model weights were saved. Training was terminated upon detection of overfitting (validation loss increase).

4.3 DUS Parameter Integration

4.3.1 Feature Extraction—17 Physical Parameters

The Distinctiveness, Uniformity, and Stability (DUS) parameters are standard morphological descriptors used in agricultural research to characterise seed varieties. Our model was extended to extract 17 physical parameters from each seed image using image processing techniques. These parameters include:

- Seed length and width
- Aspect ratio (length/width)
- Perimeter and area
- Circularity and solidity
- Eccentricity
- Moments of inertia
- Hu moments (7-dimensional shape descriptor invariant to rotation, scale, and translation)

These features provide a direct mapping from visual characteristics to standardised agricultural descriptors, enabling variety matching and quality assessment beyond class labels.

4.3.2 Pixel-to-Physical Unit Conversion

To convert pixel-based measurements to real-world physical units (millimetres), a calibration step was performed. A reference object of known size (calibration ruler) was included in the imaging setup. The pixel-to-millimetre conversion factor was computed as:

$$\text{scale} = \frac{\text{reference_length_mm}}{\text{reference_length_pixels}} \quad (4.2)$$

All extracted measurements were then scaled using this factor, ensuring that reported physical parameters are in standard units independent of camera resolution or imaging distance.

4.3.3 Variety Matching (12 Reference Varieties)

A reference database of 12 authoritative paddy seed varieties was compiled from ICAR documentation. Each reference variety is represented by median values of the 17 physical parameters computed from multiple certified images. When processing a new image, the extracted parameters are compared against all 12 reference varieties using Euclidean distance:

$$d_i = \sqrt{\sum_{j=1}^{17} (p_j - r_{i,j})^2} \quad (4.3)$$

where p_j is the j -th extracted parameter, $r_{i,j}$ is the j -th median reference value for variety i , and d_i is the distance to variety i . The variety with the smallest distance is reported as the predicted variety.

4.4 Training Pipeline

4.4.1 Phase 1—Classification Head Training (Epochs 1–8)

During Phase 1, the backbone weights were frozen and only the custom classification head was optimised. This phase lasted 8 epochs with a learning rate of 0.001. The objective was to rapidly converge the head weights to a reasonable initialisation before fine-tuning the entire network.

The training batch size was set to 32 samples, and the loss function was Cross-Entropy with label smoothing ($\epsilon = 0.1$). Validation performance was monitored at the end of each epoch to ensure stable convergence and to detect early signs of overfitting.

4.4.2 Phase 2—Full Fine-Tuning (Epochs 9–15)

During Phase 2 (epochs 9–15), all backbone layers were unfrozen and the entire network was jointly optimised. The learning rate was reduced by a factor of 10 to 0.0001, a common strategy to avoid catastrophic forgetting of pre-trained weights. This phase refined the backbone features to be better suited to the seed classification task.

Training was halted at epoch 15 when early stopping criteria were triggered (validation loss increased, indicating onset of overfitting). The best model weights from the entire training run were restored and used for final evaluation.

4.4.3 Hyperparameter Configuration and Optimiser

Table 4.4 summarises the complete hyperparameter configuration used during training.

Table 4.4: Hyperparameter Configuration and Training Settings

Hyperparameter	Value	Notes
Optimiser	AdamW	Decoupled weight decay
Phase 1 Learning Rate	0.001	Head-only training
Phase 2 Learning Rate	0.0001	Full fine-tuning (10 $\times$ reduction)
Weight Decay (L2)	0.0001	Applied to all trainable weights
Batch Size	32	GPU-optimised mini-batch
Loss Function	Cross-Entropy + Label Smoothing	$\epsilon = 0.1$, prevents overconfidence
LR Scheduler	StepLR	step=8, $\gamma = 0.1$
Gradient Clipping	Max norm = 1.0	Prevents exploding gradients
Early Stopping Patience	10 epochs	Monitors validation loss
Input Resolution	224 $\times$ 224 pixels	ResNet50 standard input
Total Epochs	15	Halted at overfitting onset

The AdamW optimiser was selected for its adaptive learning rate properties and decoupled weight decay, which provides better generalisation compared to standard L2 regularisation in the original Adam implementation.

4.4.4 Model Export Formats

Upon completion of training, the final model was exported in multiple formats to support deployment across different runtime environments. Table 4.5 lists the export formats and their intended use cases.

Table 4.5: Model Export Formats

Format	Extension	Size	Use Case
PyTorch Native	.pth	~98 MB	Primary format; FastAPI serving
TorchScript	.pt	~97 MB	Production deployment; C++ runtimes
ONNX	.onnx	~98 MB	Cross-platform; non-PyTorch frameworks
Configuration	.json	~2 KB	Model metadata, labels, hyperparameters
DUS Reference DB	.csv	<1 MB	12 reference varieties for matching

4.5 Model Deployment

Following training, the model was integrated into a production-ready web service and deployed to Google Cloud Platform (GCP). The deployment architecture is designed for scalability, reproducibility, and continuous delivery, ensuring that the model is accessible as a live API endpoint for downstream applications and testing.

4.5.1 Serving Infrastructure

The trained model (`.pth`) is loaded and served via FastAPI, a high-performance Python web framework. A single API endpoint accepts an image as input, runs it through the ResNet50 inference pipeline, and returns the quality prediction, confidence scores, and extracted physical parameters in a structured JSON response. The inference flow is as follows:

1. **Request**—FastAPI endpoint receives the image payload.
2. **Preprocessing**—Image is resized to 224×224 pixels and normalised using ImageNet statistics.
3. **Model Inference**—ResNet50 processes the image and produces class logits.
4. **Post-processing**—Softmax converts logits to probabilities; DUS parameters are extracted and variety matching is performed.
5. **Response**—Structured JSON containing prediction, confidence, physical parameters, DUS classification, and variety matches.

4.5.2 Containerisation and CI/CD Pipeline

The entire application—model weights, FastAPI server, and all dependencies—is packaged inside a Docker container, ensuring a consistent and reproducible runtime environment across development and production. Deployment is fully automated through a GitHub Actions CI/CD pipeline that executes the following steps on every push to the main branch:

1. **Build**—Docker image is built from the project Dockerfile.
2. **Push**—The built image is pushed to GCP Artifact Registry for versioned storage.

3. **Deploy**—The new image is deployed to GCP Cloud Run, which automatically provisions a serverless, auto-scaling container instance.

GCP Cloud Run was selected for its serverless auto-scaling model, which eliminates idle resource costs while handling concurrent inference requests elastically. The live API endpoint is publicly accessible at:

`https://resnet-api-297419987783.asia-south1.run.app/`

4.6 Model Comparison and Selection

To justify the selection of ResNet50 as the final architecture, a comparative evaluation of four widely used convolutional neural network architectures was conducted on the same dataset, training split, and hyperparameter configuration. The models evaluated were ResNet50, MobileNetV2, DenseNet121, and GoogleNet. Table 4.6 presents the results.

Table 4.6: Architecture Comparison on Test Set

Model	Accuracy	Precision	Recall	F1 Score	Time (min)
ResNet50	99.57%	99.57%	99.57%	99.57%	16.4
DenseNet121	99.57%	99.57%	99.57%	99.57%	15.0
GoogleNet	99.57%	99.57%	99.57%	99.57%	12.2
MobileNetV2	99.13%	99.15%	99.13%	99.13%	8.5

ResNet50, DenseNet121, and GoogleNet achieved identical peak accuracy of 99.57% on the test set, demonstrating that the dataset is well within the representational capacity of all three architectures. MobileNetV2 achieved a slightly lower accuracy of 99.13% but completed training in only 8.5 minutes—approximately half the time of ResNet50—making it a viable alternative in resource-constrained deployment scenarios.

ResNet50 was selected as the primary model for three reasons:

1. Its residual skip connections provide well-understood gradient flow properties that simplify fine-tuning behaviour analysis.
2. It has the broadest ecosystem support across deployment frameworks (PyTorch, TorchScript, ONNX, TensorRT).

3. Its architecture aligns with the majority of published benchmarks in agricultural image classification, making our results directly comparable to prior work.

The 4.4-minute training overhead compared to GoogleNet is a negligible cost given that training is performed only once.

4.7 Final Model Performance Summary

The final ResNet50 model was evaluated on the held-out test set following the completion of Phase 2 training. Table 4.7 presents the overall performance metrics, and Table 4.8 summarises key training behaviour observed across both phases.

Table 4.7: Final Test Set Performance—ResNet50

Metric	Value	Status
Test Accuracy	99.57%	Exceeds industry benchmark (85–90%)
Precision	~0.99	Excellent—minimal false positives
Recall	~0.99	Excellent—minimal false negatives
F1 Score	~0.99	Excellent—strong harmonic balance
Train-Validation Gap	<5%	Excellent generalisation; overfitting controlled

Table 4.8: Training Behaviour Summary

Observation	Detail
Initial Convergence	Rapid accuracy improvement from ~86% in Epoch 1 to near-peak within 5 epochs
Loss Trajectory	Decreased from ~1.0 to near 0.0 across 15 epochs
Mid-training Stability	Stable, consistent learning after mid-training phase with no erratic oscillations
Overfitting Control	Train-validation gap remained below the critical 5% threshold throughout
Training Stopped at	Epoch 15—onset of overfitting detected; best weights preserved

The final accuracy of 99.57% significantly exceeds the typical industry benchmark of 85–90% for automated seed quality classification systems. The near-perfect precision, recall, and F1 score across all three quality classes confirm that the model can reliably distinguish healthy seeds from damaged and fake/low quality seeds—a critical requirement for practical agricultural deployment at Manipur Technical University and partner ICAR facilities.

Chapter 5

Model Serving API

Modern machine learning systems increasingly rely on scalable and production-ready inference services capable of delivering predictions with low latency, high reliability, and secure deployment practices. While model training is computationally intensive and often performed offline, inference systems operate in real-time environments where responsiveness, scalability, and operational stability are critical. Consequently, the deployment layer of a machine learning application becomes equally important as the model itself.

This chapter presents the architecture and implementation details of the proposed Model Serving API developed using FastAPI and Uvicorn. The chapter discusses the complete inference workflow, including request handling, image pre-processing, prediction generation, confidence scoring, and latency measurement. In addition, deployment-related considerations such as containerization, environment management, observability, and security hardening are examined to establish a production-oriented inference architecture.

5.1 API Architecture Overview

5.1.1 FastAPI and Uvicorn Serving Stack

The inference service is implemented using the FastAPI framework together with the Uvicorn ASGI server. FastAPI was selected due to its high performance, asynchronous request handling capability, automatic request validation, and native OpenAPI schema generation. Uvicorn provides an efficient ASGI-compatible runtime capable of serving concurrent inference requests with low overhead.

The serving stack follows a modular architecture in which the application layer, pre-

processing pipeline, model inference engine, and response serialization components remain logically separated. Such separation improves maintainability, extensibility, and debugging efficiency.

The system exposes two primary REST endpoints:

- `POST /predict` — performs image classification inference.
- `GET /health` — provides liveness and readiness verification.

The overall architecture enables lightweight deployment while maintaining compatibility with container orchestration platforms and cloud-native deployment environments.

5.1.2 End-to-End Request Flow

The inference lifecycle begins when a client uploads an image to the prediction endpoint. The request passes through multiple processing stages before the final prediction response is returned.

The complete request flow can be summarized as follows:

1. The client uploads an image through the `/predict` endpoint.
2. FastAPI validates the incoming request and verifies payload constraints.
3. The image is decoded and transformed into a model-compatible tensor representation.
4. The trained neural network executes inference on the processed tensor.
5. Prediction probabilities are computed and ranked.
6. The system generates the final JSON response containing prediction metadata.

This architecture ensures clear separation between transport-level request handling and model execution logic, thereby improving maintainability and operational reliability.

5.1.3 Concurrency and Performance Considerations

Serving machine learning models in real-time environments introduces several performance-related challenges, particularly when handling concurrent requests. The inference service therefore incorporates several optimization strategies.

- **Worker-based execution:** Multiple worker processes may be configured depending on available CPU resources and workload characteristics.
- **Model warm loading:** Model weights are loaded during application startup to eliminate cold-start latency during the first prediction request.
- **Asynchronous request handling:** FastAPI supports asynchronous endpoint execution, allowing efficient management of concurrent I/O-bound operations.
- **Micro-batching support:** Optional request batching may improve throughput under high-load conditions.

The balance between latency and throughput depends on the deployment environment, hardware configuration, and traffic pattern of the inference workload.

5.2 Inference Pipeline

The inference pipeline transforms raw image input into a structured prediction response. This pipeline includes preprocessing, tensor transformation, model execution, confidence estimation, and response generation.

5.2.1 Input Preprocessing and Normalization

5.2.1.1 Accepted Input Formats

The API accepts image data using multipart form uploads. Commonly supported image formats include:

- JPEG
- PNG
- JPG

The uploaded image is transmitted using the `multipart/form-data` content type through the `image` request field.

5.2.1.2 Preprocessing Workflow

Before inference can be executed, the uploaded image must undergo several preprocessing operations to ensure consistency with the training pipeline used during model development.

The preprocessing sequence consists of:

1. Decoding raw image bytes into memory.
2. Converting the image into RGB color format.
3. Resizing the image to the required input dimensions.
4. Transforming the image into tensor representation.
5. Scaling pixel values into normalized numerical ranges.
6. Applying mean and standard deviation normalization.
7. Expanding tensor dimensions to include batch size information.

For image classification models trained using convolutional neural networks, maintaining preprocessing consistency between training and inference stages is essential for prediction accuracy.

5.2.1.3 Input Validation and Error Handling

Robust validation mechanisms are necessary to prevent malformed requests from causing instability within the inference system.

The service validates:

- File existence
- Supported content types
- Upload size constraints
- Image decoding success
- Safe image dimensions

If validation fails, the API returns appropriate client-side error responses using structured JSON messages. This approach improves client-side debugging while preventing unhandled server exceptions.

5.2.2 Prediction and Confidence Scoring

5.2.2.1 Inference Execution

Once preprocessing is completed, the tensor representation is forwarded to the trained neural network for prediction.

The model generates either:

- Raw logits
- Probability distributions

If logits are produced, a softmax transformation is applied to obtain normalized probability scores.

5.2.2.2 Top-1 Prediction

The class associated with the highest probability score is selected as the final prediction.

The API response may contain:

- Predicted class label
- Predicted class index
- Confidence probability

This information enables downstream systems to interpret prediction reliability and classification outcomes effectively.

5.2.2.3 Confidence Calibration

Prediction confidence plays an important role in evaluating model reliability. In production systems, poorly calibrated confidence values may lead to overconfident predictions even when the model is uncertain.

To improve prediction reliability, the system may optionally incorporate:

- Temperature scaling
- Top-k prediction outputs

- Confidence thresholds
- Uncertainty-aware rejection mechanisms

Such techniques improve interpretability and reduce the likelihood of unreliable automated decisions.

5.2.3 Latency Measurement

5.2.3.1 Inference Timing

Inference latency represents the total processing time required to generate a prediction response. Accurate latency measurement is important for evaluating deployment efficiency and maintaining service-level objectives.

The latency timer begins before preprocessing execution and stops after post-processing and response assembly are completed.

Latency values are reported in milliseconds using the `inference_ms` response field.

5.2.3.2 Measurement Methodology

The implementation uses high-resolution monotonic clocks for reliable timing measurements.

In Python-based systems, recommended timing functions include:

- `time.perf_counter()`
- `time.perf_counter_ns()`

Monotonic timing mechanisms avoid inaccuracies caused by system clock adjustments.

5.2.3.3 Operational Importance of Latency Metrics

Latency monitoring provides valuable operational insights for production environments.

These metrics assist in:

- Performance benchmarking

- Capacity planning
- Bottleneck identification
- Regression analysis
- SLA monitoring

Continuous latency tracking is particularly important when updating models or modifying preprocessing pipelines.

5.3 API Specification

5.3.1 POST /predict Endpoint

5.3.1.1 Purpose

The prediction endpoint performs image classification inference and returns prediction metadata in JSON format.

5.3.1.2 Request Format

5.3.1.2.1 HTTP Method

- POST

5.3.1.2.2 Endpoint Path

- /predict

5.3.1.2.3 Content Type

- multipart/form-data

Field	Type	Description
image	File	Uploaded image for inference

Table 5.1: Prediction endpoint request parameters

5.3.1.2.4 Request Parameters

5.3.1.3 Response Format

A successful prediction request returns:

```
{
  "predicted_label": "class_name",
  "confidence": 0.987,
  "inference_ms": 12.4
}
```

Optional response metadata may include:

- Predicted class index
- Top-k predictions
- Model version
- Request identifier

5.3.1.4 Error Responses

The API returns structured error responses for invalid requests.

Status Code	Description
400	Invalid request payload or missing image
413	Uploaded file exceeds size limit
415	Unsupported media type
500	Internal server-side exception
503	Model unavailable or initialization failure

Table 5.2: Prediction endpoint error responses

5.3.2 GET /health Endpoint

5.3.2.1 Purpose

The health endpoint provides operational monitoring support for deployment platforms, orchestrators, and uptime monitoring systems.

5.3.2.2 Request Specification

- Method: GET
- Path: /health

5.3.2.3 Response Format

The endpoint returns a lightweight JSON response:

```
{  
  "status": "ok"  
}
```

Additional readiness information may also be included:

```
{  
  "status": "ok",  
  "model_loaded": true,  
  "version": "1.0.0"  
}
```

5.3.2.4 Liveness and Readiness Checks

Containerized deployment environments frequently distinguish between:

- **Liveness checks** — verifies whether the process is operational.
- **Readiness checks** — verifies whether the model is fully initialized and capable of serving requests.

These mechanisms improve deployment stability and automated recovery behavior.

5.4 Containerization and Deployment

5.4.1 Docker Multi-Stage Build Architecture

Containerization simplifies deployment consistency across development, testing, and production environments.

The proposed deployment architecture uses a multi-stage Docker build process to minimize final image size and reduce unnecessary runtime dependencies.

5.4.1.1 Builder Stage

The builder stage performs:

- Dependency installation
- Wheel generation
- Environment preparation
- Optional testing and validation

5.4.1.2 Runtime Stage

The runtime stage contains only the components necessary for serving inference requests.

The runtime image includes:

- Application source code
- Installed dependencies
- Model artifacts
- Runtime configuration

The container executes using a non-root user to reduce security risks.

5.4.2 Deployment Considerations

Several operational considerations influence deployment efficiency:

- Dependency version pinning
- Slim container base images
- Non-root execution
- Resource limit enforcement
- Horizontal scalability

These practices improve reproducibility, reliability, and operational security.

5.5 Configuration and Environment Management

5.5.1 Environment-Based Configuration

The system follows a 12-factor application approach in which runtime behavior is controlled through environment variables.

Common configuration parameters include:

- `MODEL_PATH`
- `MODEL_VERSION`
- `LOG_LEVEL`
- `MAX_UPLOAD_MB`
- `ALLOWED_ORIGINS`

Separating configuration from source code improves portability and deployment flexibility.

5.5.2 Secrets Management

Sensitive information should never be hardcoded within application source files.

Recommended approaches include:

- Environment-based secret injection
- Cloud secret management systems
- Kubernetes Secrets
- Role-based access control

Secure secret management reduces the likelihood of credential leakage and unauthorized access.

5.6 Security Considerations

5.6.1 Threat Model

Inference APIs exposed to public or semi-public environments are vulnerable to several operational and security threats.

Potential attack vectors include:

- Malformed image payloads
- Denial-of-service attacks
- Excessive resource consumption
- Dependency vulnerabilities
- Container privilege escalation
- Information leakage through logs

Therefore, security hardening must be integrated throughout the deployment lifecycle.

5.6.2 Input Hardening

The API enforces several defensive validation mechanisms:

- Upload size restrictions
- Content type validation
- Safe image dimension limits
- Structured error responses

These safeguards reduce the likelihood of memory exhaustion and malicious payload execution.

5.6.3 Transport Security

All production deployments should enforce HTTPS/TLS encryption to protect data transmitted between clients and the inference server.

TLS termination may be handled through:

- Reverse proxies
- Ingress controllers
- Cloud load balancers

Encrypted transport channels prevent interception and tampering of inference requests.

5.6.4 Rate Limiting and Abuse Prevention

To protect system availability, the deployment may incorporate:

- Per-client request throttling
- Token-based quotas
- Request queuing
- Circuit breaker mechanisms

These controls reduce the impact of malicious traffic spikes and abusive usage patterns.

5.6.5 Logging and Observability

Operational observability is necessary for diagnosing failures and monitoring service health.

The system should record:

- Request identifiers
- Inference latency
- Error statistics
- Deployment version metadata

However, sensitive user data and raw image payloads should not be stored in logs.

5.7 Summary

This chapter presented the architecture and implementation of the proposed Model Serving API developed using FastAPI and Uvicorn. The discussion covered the complete inference pipeline, including image preprocessing, tensor normalization, prediction generation, confidence estimation, and latency measurement. The chapter also examined API endpoint specifications, containerized deployment using Docker multi-stage builds, environment-based configuration management, and security hardening practices. Collectively, these components establish a scalable and production-oriented inference infrastructure suitable for modern machine learning deployment environments.

Chapter 6

Frontend System Architecture and Implementation

6.1 Architecture and Design Principles

The AgriVerify frontend is implemented as a Next.js 15 application following a layered component-service architecture [21]. Rather than concentrating business logic inside route pages, the design distributes responsibilities across four distinct layers: presentation, query, service, and transport. Each layer has a single, well-defined concern, which keeps individual files small and makes the dependency chain easy to follow.

At the topmost layer, route page components remain thin. They declare what data they need and how it should be displayed, but they delegate all asynchronous behavior to custom hooks. The hook layer wraps TanStack Query [22] mutations and queries, exposing loading, error, and data states to the page. Beneath the hooks, domain service modules group backend-facing functions by subject area, such as authentication, verification, complaints, and administration. Finally, a shared Axios [23] client handles token injection and refresh-token recovery, forwarding every request to a Next.js proxy route that relays it to the ASP.NET REST backend.

This layering produces several practical benefits. Testing any individual layer does not require instantiating the full component tree. New features can extend an existing service module without modifying page components. Authentication changes propagate through a single Axios interceptor rather than being scattered across individual fetch calls. The architecture is therefore both extensible and straightforward to audit.

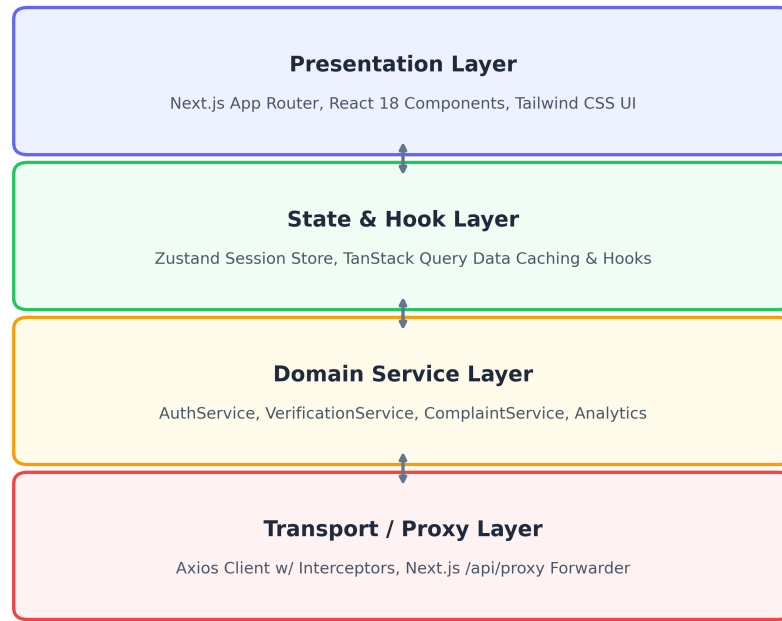


Figure 6.1: Layered frontend architecture and data flow for the AgriVerify application

6.2 Technology Stack

Table 6.1 lists the primary libraries and frameworks used in the frontend. The choices reflect a preference for well-maintained, type-safe tooling that integrates cleanly with the Next.js App Router [24].

Typography is handled through Next.js native font loading. The root layout loads DM Sans for body text and Sora for display headings via `next/font/google`, which injects both families as CSS variables. The Tailwind configuration then maps these variables into semantic font families, giving the application a consistent typographic hierarchy without any runtime font-fetch overhead in the browser.

Notably absent from the dependency list are a dedicated form validation library and a charting package. Forms across the application are handled with controlled inputs and manual validation checks, while analytics screens present data through metric cards, progress bars, and tabular summaries rather than graphical charts. This keeps the bundle lean and reduces the number of abstraction layers that must be traced when debugging a form submission or a dashboard figure.

Table 6.1: Technology stack used by the AgriVerify frontend

Component	Version	Role in the application
Next.js	15.2.4	Application framework, App Router, and server-side proxy route
React	19.2.5	Component rendering and local UI state
TypeScript	5.7.3	Static typing across UI, services, hooks, and shared models
Tailwind CSS	3.4.17	Utility-first styling and design-token application
Axios	1.7.9	HTTP client with request and response interceptors
TanStack Query	5.59.0	Server-state caching, invalidation, and query lifecycle management
Zustand	5.0.1	Lightweight global store for authentication state
Framer Motion	12.38.0	Page transitions and animated state changes
Lucide React	1.14.0	Consistent icon library
react-webcam	7.2.0	Camera capture interface for the public verification scanner

6.3 Folder Structure and Repository Layout

The source tree is rooted at `src/` and partitioned into seven top-level directories. Table 6.2 describes the responsibility of each area.

Within `src/app/`, the Next.js App Router organizes routes into four groups: `(auth)`, `(farmer)`, `(officer)`, and `(admin)`. Route group parentheses do not appear in the public URL, so a file at `src/app/(farmer)/farmer/dashboard/page.tsx` resolves cleanly to `/farmer/dashboard`. A catch-all proxy handler at `src/app/api/proxy/` relays every browser request to the ASP.NET backend, keeping the backend base URL out of the browser environment entirely.

6.4 Routing System

6.4.1 App Router and Route Groups

AgriVerify uses the Next.js App Router exclusively. Each route is defined by a `page.tsx` file, and layouts at each level of the hierarchy inject shared navigation, providers, and access guards. Route groups separate the four major access domains — public, authentication, farmer, and administration — without introducing extra path segments.

Table 6.2: Directory responsibilities in the AgriVerify frontend repository

Directory	Responsibility
<code>src/app/</code>	Next.js routes, layouts, route groups, and the API proxy handler
<code>src/components/</code>	Shared UI primitives and feature components such as camera capture, verification results, and branding elements
<code>src/hooks/</code>	Custom hooks built on TanStack Query for authentication, complaints, verification, analytics, administration, and reference data
<code>src/services/</code>	Backend-facing API functions grouped by domain
<code>src/lib/</code>	Utility functions including error normalization and role-to-route mapping
<code>src/store/</code>	Zustand store for authentication state
<code>src/types/</code>	Shared TypeScript interfaces and response models

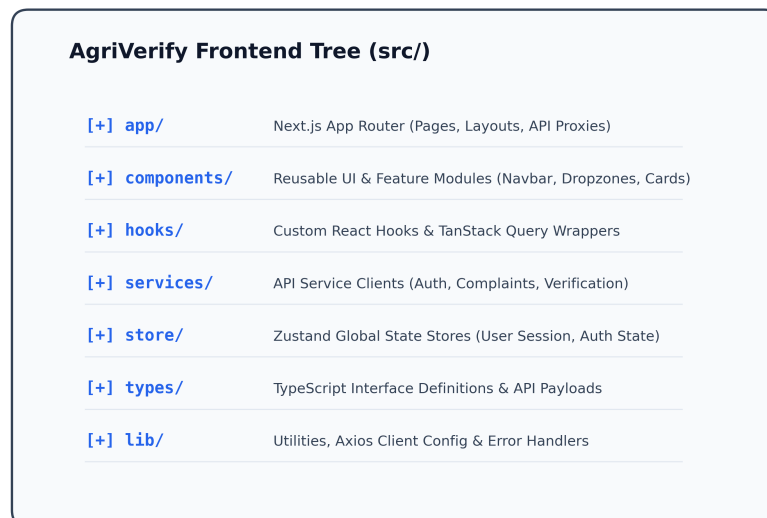


Figure 6.2: Logical folder organization of the AgriVerify frontend source tree

The visible URL space is as follows:

- / — public landing page;
- /verify — anonymous seed-verification scanner;
- /login and /register — authentication entry points;
- /farmer/dashboard, /farmer/verify, /farmer/history, /farmer/complaints — farmer portal pages;
- /officer/dashboard, /officer/complaints — officer complaint queue;
- /admin/dashboard, /admin/officers, /admin/farmers, /admin/analytics, /admin/audit-logs — administrative management views.

The root layout applies global metadata, imports the Tailwind stylesheet, and wraps the entire application in the `Providers` component, which instantiates the TanStack Query client, mounts the global toast notification system, and exposes React Query Devtools in development mode.

6.4.2 Role-Based Access Control

Route protection is enforced through a shared `RoleGuard` component rather than a Next.js middleware file. Every role-specific layout wraps its children in `RoleGuard`, centralizing the access check so that it is not duplicated across individual pages. On mount, the guard reads the effective token and role from the Zustand store, falling back to `localStorage` when the in-memory store has not yet been hydrated. If no valid token is found, the user is redirected to `/login`. If a token is present but the role does not match the allowed role for the current layout, the guard resolves the user's correct dashboard path via `getDashboardPathByRole` and redirects accordingly.

```
1 export default function RoleGuard({ allowedRole, children }:  
  RoleGuardProps) {  
2   const router = useRouter()  
3   const pathname = usePathname()  
4   const token = useAuthStore((s) => s.token)  
5   const role = useAuthStore((s) => s.role)  
6  
7   const localToken = typeof window !== 'undefined'  
8     ? window.localStorage.getItem('accessToken') : null  
9   const localRole = typeof window !== 'undefined'
```

```

10     ? JSON.parse(window.localStorage.getItem('user')) ?? 'null')?.role
11     ?? null
12     : null
13
14     const effectiveToken = token ?? localToken
15     const effectiveRole = role ?? localRole
16
17     useEffect(() => {
18         if (!effectiveToken) { router.replace('/login'); return }
19         if (effectiveRole !== allowedRole) {
20             const fallback = getDashboardPathByRole(effectiveRole)
21             if (pathname !== fallback) router.replace(fallback)
22         }
23     }, [allowedRole, effectiveRole, effectiveToken, pathname, router])
24
25     if (!isAllowed) return <div>Checking access...</div>
26     return <>{children}</>

```

Listing 6.1: RoleGuard: client-side route enforcement for role-partitioned layouts

The role-to-dashboard mapping is maintained in `src/lib/authRouting.ts`: Admin resolves to `/admin/dashboard`, Officer to `/officer/dashboard`, and all remaining authenticated users fall back to `/farmer/dashboard`. Centralizing this mapping prevents route drift when new role-specific pages are introduced.

6.4.2.1 Farmer Portal — Verification and Complaint Workflows

The farmer portal is the primary interface for registered farmers. After authentication, farmers are directed to a dashboard that presents headline metrics: total verifications performed, the proportion of genuine versus suspect results, and active complaint counts. From the dashboard, a farmer can initiate a new seed verification by uploading front and back photographs of a seed packet along with the crop type and district metadata. Completed verifications are archived in a paginated history view, where each entry can be expanded to reveal the full analysis report including confidence scores, image previews, and batch-risk indicators.

The complaint module allows farmers to raise a formal grievance when a verification result conflicts with field experience. A complaint form collects a descriptive summary, severity rating, and the associated verification record. Submitted complaints are tracked in a status-aware list that reflects officer assignments and resolution updates in near real time through React Query polling.

6.4.2.2 Officer Portal — Complaint Processing Queue

Officers access a focused work queue that surfaces all complaints assigned to the current officer along with unassigned complaints in their jurisdiction. The queue table supports filtering by status and district, allowing officers to prioritize pending items efficiently. Selecting a complaint row opens a detail panel that displays the originating verification images, the farmer’s narrative, and the full audit history for that complaint record. Officers update the complaint status through a controlled dropdown, and each transition triggers an optimistic UI update followed by a cache invalidation to keep the list consistent.

6.4.2.3 Admin Portal — Analytics and Management

The administrative dashboard aggregates platform-wide activity. Metric cards at the top of the screen display total verifications, active complaints, registered farmers, and officer counts. Below the metrics, tabular views support officer management — including creation, profile updates, and deactivation — and farmer account oversight, where an administrator can activate or suspend accounts and inspect onboarding details. An analytics section presents verification and complaint metrics across user-selected date ranges. Audit logs provide a chronological record of all state-changing operations for compliance and accountability purposes.

6.4.2.4 Public Verification — Anonymous Scanner

The public verification route at `/verify` provides an unauthenticated experience suitable for farmers who have not yet registered or who wish to perform a quick scan in the field. The `CameraCapture` component activates the device camera through `react-webcam` and guides the user through capturing both faces of a seed packet. The captured images are submitted to the `/verification/verify-anonymous` endpoint without any authentication header. The resulting `VerificationResult` panel renders the classification label, confidence level, visual match summary, and a recommendation to file a complaint or contact the agriculture department if the result is suspect. A separate placeholder route at `/farmer/verify/anonymous` redirects to `/verify` to avoid duplicating the scanner implementation.

6.5 Authentication Flow

6.5.1 Token Storage Strategy and Zustand Auth Store

AgriVerify uses a token-based authentication model. Upon a successful login or registration, the ASP.NET backend returns an access token, a refresh token, expiry metadata, and a serialized user profile. The frontend stores all four values in `localStorage` and simultaneously writes them into the Zustand auth store, making the session immediately available to the Axios request interceptor and all authenticated components.

The Zustand store exposes four state fields — `user`, `token`, `role`, and `isAuthenticated` — and two actions: `setAuth` and `clearAuth`. During initial client-side hydration, the store reads `localStorage` to restore the prior session without requiring a fresh login on page reload.

```
1 export const useAuthStore = create<AuthStore>((set) => ({
2   ...getInitialAuthState(),
3   setAuth: (token, refreshToken, user) => {
4     if (typeof window !== 'undefined') {
5       window.localStorage.setItem('accessToken', token)
6       window.localStorage.setItem('refreshToken', refreshToken)
7       window.localStorage.setItem('user', JSON.stringify(user
8     ))
9   }
10   set({ user, token, role: user.role ?? null, isAuthenticated: true
11   })
12 },
13 clearAuth: () => {
14   if (typeof window !== 'undefined') {
15     window.localStorage.removeItem('accessToken')
16     window.localStorage.removeItem('refreshToken')
17     window.localStorage.removeItem('user')
18   }
19   set({ user: null, token: null, role: null, isAuthenticated: false
20   })
21 },
22 })))
```

Listing 6.2: Zustand authentication store with browser-storage persistence

Storing tokens in `localStorage` is straightforward to implement and survives page reloads, but it exposes the session to JavaScript-accessible storage. A cross-site scripting compromise would therefore be able to read the token directly. For a production hardening pass, replacing `localStorage` with `httpOnly` cookies

and a CSRF-aware session design would eliminate this attack surface [25].

6.5.2 Refresh Token Recovery (Axios 401 Interceptor)

When the access token expires, the backend returns a 401 `Unauthorized` response. The Axios response interceptor detects this condition and, provided the request has not already been retried, posts the stored refresh token to `/api/proxy/auth/refresh`. If the backend accepts the refresh token, the interceptor writes the new access and refresh tokens back to `localStorage` and to the Zustand store, then reissues the original request transparently. If the refresh call itself fails — for example because the refresh token has also expired — the interceptor calls `clearAuth`, clears all stored tokens, and redirects the browser to `/login`. This design means that individual components never need to implement token-expiry logic; the interceptor handles it universally for every request in the application.

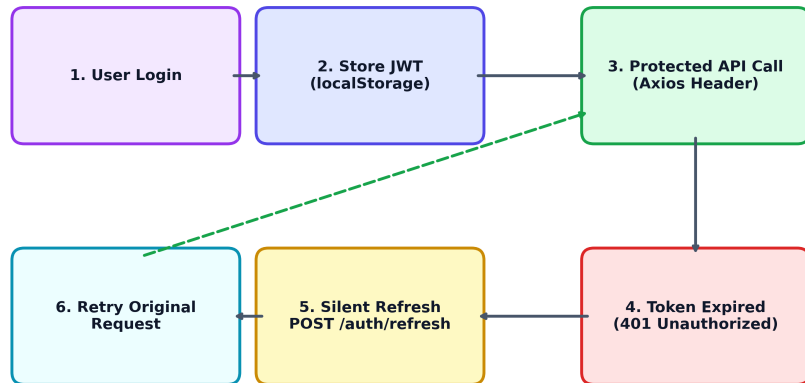


Figure 6.3: Authentication lifecycle including token persistence and refresh recovery

6.6 API Integration

6.6.1 Axios Client, Request Interceptor, and Proxy Route

All HTTP communication is routed through a single Axios instance whose `baseUrl` is set to `/api/proxy`. A request interceptor reads the access token from `localStorage` and attaches it as a `Bearer` authorization header on every outgoing request. Because the interceptor operates at the transport layer rather than within individual

service functions, token injection is guaranteed regardless of which service module initiates the call.

```
1 export const apiClient = axios.create({
2   baseURL: '/api/proxy',
3   headers: { 'Content-Type': 'application/json' },
4 })
5
6 apiClient.interceptors.request.use((config) => {
7   if (typeof window !== 'undefined') {
8     const token = window.localStorage.getItem('accessToken')
9     if (token) config.headers.Authorization = `Bearer ${token}`
10  }
11  return config
12 })
```

Listing 6.3: Centralized Axios client with bearer-token injection

The Next.js proxy route at `src/app/api/proxy/[...path]/route.ts` receives the proxied request, reconstructs the target URL by prefixing the path with `/api/v1/`, and forwards the call to the ASP.NET backend whose address is stored in the server-only environment variable `BACKEND_API_URL`. Authorization and content-type headers are forwarded verbatim. Multipart form-data payloads — used for seed-image uploads — are piped through as raw request bodies without re-parsing, which preserves the boundary delimiters required by the backend’s file-upload handler. JSON payloads for POST and PUT requests are deserialized and re-serialized so that the proxy can log or inspect them if needed. Response status codes and headers from the ASP.NET backend are returned to the browser unchanged, ensuring that HTTP semantics such as cache-control and content-type are preserved end to end.

6.6.2 Endpoint Summary

The service layer exposes grouped API functions that map to the paths listed in Tables 6.3–6.6. All paths are relative to `/api/proxy` in the browser; the proxy prepends `/api/v1/` before forwarding to the backend.

6.6.3 State Management (TanStack Query and Zustand)

The application separates server state from client state [22]. TanStack Query owns all data that originates from the backend: it fetches, caches, invalidates, and refetches records in response to mutations and stale-time expiry. Zustand owns all

Table 6.3: Authentication API endpoints

Function	Method	Path and notes
login	POST	/auth/login — farmer email and password; public.
officerLogin	POST	/auth/officer-login — officer ID and OTP; public.
register	POST	/auth/register — new farmer account; public.
refreshToken	POST	/auth/refresh — token renewal; used by 401 interceptor.
logout	POST	/auth/logout — bearer token required.
getMe	GET	/auth/me — current user profile; bearer token required.

Table 6.4: Seed verification API endpoints

Function	Method	Path and notes
uploadVerification	POST	/verification/upload — multipart; front and back images, crop, district; authenticated.
verifyAnonymous	POST	/verification/verify-anonymous — multipart; front and back images; public.
getVerificationHistory	GET	/verification/history — paginated; authenticated.
getVerification	GET	/verification/{id} — single record; authenticated.
getVerificationStatistics	GET	/verification/statistics — dashboard metrics; authenticated.

Table 6.5: Complaint management API endpoints

Function	Method	Path and notes
<code>createComplaint</code>	POST	<code>/Complaints</code> — creates a complaint record; authenticated.
<code>getMyComplaints</code>	GET	<code>/Complaints/my-complaints</code> — paginated farmer view; authenticated.
<code>getAllComplaints</code>	GET	<code>/Complaints</code> — officer/admin filtered list; role-protected.
<code>getComplaint</code>	GET	<code>/Complaints/{id}</code> — single record; authenticated.
<code>updateComplaintStatus</code>	PUT	<code>/Complaints/{id}/status</code> — officer status transition; authenticated.
<code>assignComplaint</code>	POST	<code>/complaints/{id}/assign</code> — admin assignment; authenticated.
<code>getBatchRisks</code>	GET	<code>/Complaints/batch-risks</code> — risk summary list; authenticated.
<code>getComplaintStats</code>	GET	<code>/complaints/stats</code> — aggregate statistics; authenticated.

data that describes the current user’s session. This division means that neither library is asked to perform a role it was not designed for.

Several behavioral defaults are configured at the query client level. Window-focus refetching is disabled globally to avoid spurious reloads when farmers switch between browser tabs during field use. Reference data queries — states, districts, and crops — use an infinite stale time, since this data rarely changes and repeated fetches would add unnecessary latency on low-bandwidth connections common in rural Manipur. Dashboard queries use a moderate stale window, balancing metric freshness against the cost of each request.

Mutations follow a consistent pattern: the hook performs an optimistic update on the local cache to give immediate visual feedback, submits the change to the backend, and then invalidates the corresponding list query on success. Failure cases roll back the optimistic patch and surface an error toast. This approach keeps the UI responsive even on slow connections while guaranteeing eventual consistency with the server.

Error normalization is handled centrally by the `extractErrorMessage` utility in `src/lib`. The function inspects the rejected Axios value, extracts the `detail` or `title` field from problem-detail-shaped responses, and falls back to a generic failure message when the error is not structured. Because all hooks call this utility

Table 6.6: Administration and analytics API endpoints

Function	Method	Path and notes
getOfficers	GET	/admin/officers — paginated; authenticated.
createOfficer	POST	/admin/officers — authenticated.
updateOfficer	PUT	/admin/officers/{id} — authenticated.
deleteOfficer	DELETE	/admin/officers/{id} — authenticated.
deactivateOfficer	POST	/admin/officers/{id}/deactivate — authenticated.
getFarmers	GET	/admin/farmers — paginated; authenticated.
updateFarmerStatus	PUT	/admin/farmers/{id}/status — authenticated.
getAdminDashboard	GET	/admin/dashboard — authenticated.
getAuditLogs	GET	/admin/audit-logs — paginated; authenticated.
getSystemAnalytics	GET	/analytics/system — date-range filtered; authenticated.
getVerificationMetrics	GET	/analytics/verifications — date-range filtered; authenticated.
getComplaintMetrics	GET	/analytics/complaints — date-range filtered; authenticated.
getStates	GET	/reference/states — public reference data.
getDistricts	GET	/reference/districts — public reference data.
getCrops	GET	/reference/crops — public reference data.

before passing errors to the toast system, error messages across the application are consistent and predictable.

6.6.4 UI Component Library and Design System

The repository includes a compact local component library under `src/components/ui`. Table 6.7 lists the shared primitives, each of which wraps a native HTML element or a small portal-based container.

Table 6.7: Shared UI primitives in the AgriVerify component library

Component	Purpose	Notable detail
Button	Action control	Variant and size maps encoded as Tailwind class strings
Input	Text input	Standardized focus and disabled states
Textarea	Multi-line input	Styled consistently with Input
Select	Option selection	Context-driven wrapper around native select
Checkbox	Boolean control	Emits values through <code>onCheckedChange</code>
Slider	Range input	Used for complaint severity rating
Card	Content container	Border, shadow, and spacing defaults
Badge	Status marker	Applied to verification and complaint state labels
Table	Tabular layout	Standard wrappers with hover-row styling
Dialog	Modal overlay	Portal-based with local open state
Sheet	Side-panel overlay	Portal-based responsive drawer
Tabs	Segmented UI state	Context-driven active-tab selection
Progress	Linear progress bar	CSS transform-based fill animation
Skeleton	Loading placeholder	Pulse animation for dashboard fetch states

Feature-level components sit alongside the primitives. `AgriVerifyLogo` and `LeafDecoration` provide consistent branding across public and authentication screens. `CameraCapture` encapsulates the webcam lifecycle, including permission requests and frame capture. `VerificationResult` renders the structured analysis panel with status badge, confidence breakdown, and image preview. `RoleGuard` enforces route access as described in Section 6.4.2.

Dashboard pages follow a uniform composition pattern: metric cards for headline counters, tables for record lists, badges for status labels, dialogs for row-level detail and edit flows, and skeletons during data fetches. This consistency makes the

interface visually predictable across all three role portals and reduces the cognitive overhead for users who switch between roles.

Forms are built from controlled inputs and explicit submission handlers using `useState`. The login and registration pages use a centered card layout with icon-decorated fields, password-visibility toggling, and a remember-me checkbox. Validation is performed through a combination of HTML `required` constraints, type-specific inputs such as `type="email"`, explicit equality checks for password confirmation, and toast-based error reporting when constraints are violated.

6.6.5 Responsive Design and Accessibility Notes

The application is styled exclusively with Tailwind CSS utilities [26]. The Tailwind configuration extends the default palette with a green-toned agricultural color scheme defined in the global stylesheet. Custom shadows and border radii are mapped to semantic names, and the DM Sans and Sora font families are wired into the `fontFamily` configuration for consistent typographic hierarchy.

Layouts respond to screen size through Tailwind breakpoint prefixes. On small screens, dashboard grids collapse to a single column and tables become horizontally scrollable. On medium and large screens, metric grids expand to two or three columns, and detail panels open as side sheets rather than full-page overlays. This approach is sufficient for field use on mobile devices, which is an important consideration given that many farmers in Manipur access the platform on smartphones.

Accessibility is addressed through native HTML semantics. Form fields have associated `label` elements. Interactive elements use `button` rather than `div` with click handlers. Data grids use proper `table`, `thead`, `tbody`, `th`, and `td` elements so that screen readers can announce column headers. Portal-based overlays leverage the browser's default focus management. No custom focus-trap logic is implemented beyond what the browser provides, which is adequate for the current scope but would need extension if the application were to target users who rely heavily on keyboard navigation.

6.7 Summary

The AgriVerify frontend implements a well-structured, role-aware Next.js application suitable for production deployment in low-bandwidth agricultural environments. Its layered architecture cleanly separates presentation, asynchronous

behavior, backend communication, and authentication state. The App Router route groups produce clean, predictable URLs while centralizing access control through a single guard component. TanStack Query and Zustand provide a maintainable data layer without requiring a heavyweight global state library. The local component library and Tailwind-based design system give the UI a consistent, accessible, and responsive character across all four user roles — farmer, officer, administrator, and anonymous visitor.

Future development should prioritize migrating token storage to `httpOnly` cookies, introducing a schema-based form validation library for larger forms, and adding a dedicated charting library to enrich the analytics screens currently represented as tabular summaries.

Chapter 7

Backend System Architecture and Implementation

7.1 Architectural Overview

The AI-Based Paddy Seed Quality Assessment And Farmer Feedback System employs a comprehensive backend architecture designed according to Clean Architecture principles, ensuring optimal separation of concerns and maintainability. This design paradigm facilitates loose coupling between core business logic and external dependencies such as databases, user interfaces, and third-party AI services. By adhering to these architectural principles, the system achieves high scalability, testability, and long-term maintainability essential for agricultural technology platforms.

7.1.1 Architectural Layers

The backend infrastructure is structured into four distinct, independent layers, each serving a specific purpose within the application ecosystem:

7.1.1.1 Domain Layer

The Domain Layer forms the foundation of the system, encapsulating core business entities, value objects, enums, and domain-specific exceptions. This layer remains completely independent from any external framework or infrastructure concern, representing the pure business logic of seed verification and quality assessment. It defines what the system does from a business perspective, free from technical implementation details.

7.1.1.2 Application Layer

The Application Layer orchestrates business logic by implementing application services and defining contracts through interfaces. It manages Data Transfer Objects (DTOs) for request-response mapping, establishes validation rules, and coordinates workflows between the Domain and Infrastructure layers. This layer depends exclusively on the Domain Layer, ensuring that business rules remain isolated and reusable.

7.1.1.3 Infrastructure Layer

The Infrastructure Layer implements concrete technical solutions including database access patterns using Entity Framework Core, integrations with Azure AI services (Custom Vision, Computer Vision OCR, and OpenAI), and cloud file storage mechanisms. It provides implementations for interfaces defined in higher layers while handling all external service integrations and data persistence operations.

7.1.1.4 Presentation Layer (API Layer)

The Presentation Layer serves as the application's entry point, handling HTTP requests, managing JWT-based authentication, and exposing RESTful API endpoints. This layer converts HTTP communication into application service calls and returns structured responses to clients while maintaining security through authorization and validation middleware.

7.1.2 Architectural Diagram

Figure 7.1 illustrates the hierarchical relationship and dependency flow among architectural layers:

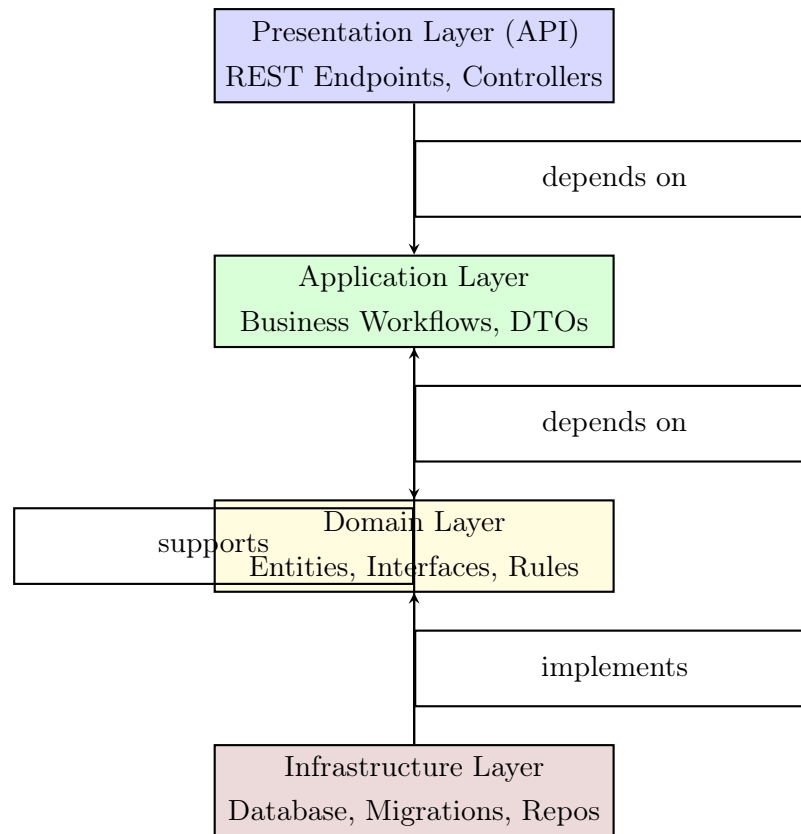


Figure 7.1: Clean Architecture Layer Hierarchy

7.2 Technology Stack

The backend system leverages a modern, enterprise-grade technology stack specifically selected for scalability, maintainability, security, and high performance in agricultural technology contexts. Table 7.1 summarizes the critical technology components:

Table 7.1: Backend Technology Stack Components

Category	Technology
Backend Framework	ASP.NET Core Web API
Programming Language	C#
Architecture Pattern	Clean Architecture
Object-Relational Mapping	Entity Framework Core
Database	SQL Server
Authentication	JWT Bearer Authentication
Identity Management	ASP.NET Core Identity
AI/ML Integration	Azure Custom Vision, Azure Computer Vision OCR, Azure OpenAI
Cloud Storage	Azure Blob Storage
Input Validation	FluentValidation
Object Mapping	AutoMapper
Logging	Serilog
API Documentation	Swagger / OpenAPI
Dependency Injection	Built-in ASP.NET Core DI Container
Caching	In-Memory Caching
Version Control	Git & GitHub

The selection of ASP.NET Core enables cross-platform deployment while maintaining enterprise-level performance characteristics. Integration with Azure services provides robust AI capabilities and cloud-native scalability, essential for processing agricultural image data and maintaining secure authentication systems.

7.3 Project Structure Analysis

7.3.1 Domain Layer Architecture

The Domain Layer encapsulates the fundamental business concepts of the Fake Seed Detection System. This layer contains core entities representing agricultural and administrative concepts, enums defining system states, interfaces specifying service contracts, and shared base classes providing common functionality.

7.3.1.1 Entities Folder — Core Business Models

The Entities folder contains the primary business objects used throughout the system. Each entity represents a key concept in the seed verification and complaint management workflow:

BaseEntity.cs Serves as the foundation for most domain entities, providing common audit properties including unique identifier, creation timestamp, modification timestamp, and audit trail information about entity lifecycle changes.

Seed.cs Represents the core seed entity containing agricultural product details necessary for verification operations.

SeedBatch.cs Models collections of seeds from a single production batch, enabling batch-level analysis and risk assessment across multiple products.

DetectionResult.cs Encapsulates the outcome of AI-based seed analysis, including confidence scores, classification results, and quality metrics.

AnalysisReport.cs Aggregates detection results and generates comprehensive reports suitable for farmer communication and administrative review.

AuditLog.cs Maintains detailed records of system operations and user actions for compliance, security auditing, and dispute resolution.

7.3.1.2 Enums Folder — System Classifications

Enums maintain consistency by restricting values to predefined sets, preventing invalid state combinations:

SeedStatus.cs Defines lifecycle states for seeds (Active, Suspended, Verified, Rejected).

QualityLevel.cs Classifies seeds into quality categories (Excellent, Good, Fair, Poor, Counterfeit) based on AI analysis.

ImageFormat.cs Enumerates supported image formats (JPG, JPEG, PNG) for seed packet analysis.

DetectionAlgorithm.cs Specifies available AI algorithms (Custom Vision, ResNet, Hybrid).

ReportType.cs Categorizes generated reports (VerificationSummary, Detailed-Analysis, RiskAssessment).

7.3.1.3 Interfaces Folder — Service Contracts

Interfaces define contracts that promote testability and loose coupling throughout the system:

IRepository<T> Generic repository interface providing standard CRUD (Create, Read, Update, Delete) operations while abstracting data access complexity from business logic.

ISeedRepository Extends the generic repository with seed-specific query operations optimized for verification workflows.

IBatchRepository Specializes repository operations for batch-level queries including batch risk assessment and historical analysis.

IDetectionService Defines contracts for image analysis and seed classification operations utilizing AI services.

IReportService Specifies interfaces for generating structured reports in various formats (PDF, Excel, JSON).

IUnitOfWork Implements the Unit of Work pattern for coordinating multiple repository operations within atomic transactions.

IAuthenticationService Defines security-related operations including user verification and access control.

7.3.1.4 Common Folder — Shared Base Classes

The Common folder provides reusable base classes reducing code duplication:

BaseEntity.cs Base class providing common audit fields (Id, CreatedAt, UpdatedAt, CreatedBy, UpdatedBy).

Result<T> Generic result wrapper enabling consistent error handling and operation status communication throughout the application.

PagedResult<T> Wrapper class supporting pagination in list operations, reducing database load for large result sets.

ApiResponse Standardizes HTTP response structure across all API endpoints.

Pagination Data transfer object containing pagination metadata (PageNumber, PageSize, TotalRecords).

7.3.1.5 Exceptions Folder — Error Handling Strategy

Custom exceptions provide semantic meaning to error conditions:

DomainException Base class for all domain-level business rule violations.

EntityNotFoundException Raised when requested entities do not exist in the system.

ValidationException Aggregates multiple validation failures with field-level details.

UnauthorizedAccessException Indicates insufficient permissions for requested operations.

InvalidOperationException Represents invalid state transitions or business logic violations.

7.3.2 Infrastructure Layer Architecture

The Infrastructure Layer implements technical solutions and external service integrations. It translates domain interfaces into concrete implementations while managing data persistence and cloud service communications.

7.3.2.1 Persistence Folder — Data Management

The Persistence folder encapsulates all database-related operations:

7.3.2.1.1 ApplicationDbContext.cs The `ApplicationDbContext` class serves as the central Entity Framework Core context, mapping C# domain entities to database tables. It manages:

- Entity-to-table mappings through `DbSet` properties
- Database change tracking and automatic timestamp management
- Soft delete functionality preserving historical records
- Relationship configurations and constraints
- Audit field population (`CreatedAt`, `UpdatedAt` automatically)

7.3.2.1.2 UnitOfWork.cs Implements the Unit of Work pattern coordinating multiple database operations as atomic transactions. When processing seed complaints with simultaneous investigation record creation, UnitOfWork ensures both operations commit together or both rollback, preventing data inconsistency.

7.3.2.1.3 Repository Implementation

Repository<T>.cs Generic repository implementing standard CRUD operations and reducing code duplication across entity-specific repositories.

SeedRepository.cs Provides seed-specific queries including batch-level searches and quality assessment lookups.

BatchRepository.cs Manages batch operations including risk analysis and historical trend queries.

DetectionResultRepository.cs Specializes in verification result retrieval and analysis queries.

7.3.2.2 Configurations Folder — Database Mapping

Entity Framework Core configuration classes define database schema, indexes, and relationships:

SeedEntityTypeConfiguration Defines table schema, column types, constraints, and indexes for optimal seed query performance.

BatchEntityTypeConfiguration Configures batch table structure with relationships to seed and result entities.

ResultEntityTypeConfiguration Maps detection result schema with confidence score storage and enum conversions.

AuditLogConfiguration Establishes audit log structure with timestamp indexing for efficient historical queries.

7.3.2.3 Migrations Folder — Schema Version Control

Database migrations enable safe schema evolution without data loss:

- Initial migrations create core tables and relationships

- Incremental migrations add features (new columns, tables, indexes) as application requirements evolve
- Migration history enables rollback to previous schema versions if necessary
- Each migration is timestamped and sequenced for deterministic application

Example migrations include:

20240101000000_InitialCreate.cs Creates foundational entities (Seed, Batch, User, DetectionResult).

20240115000000_AddDetectionTables.cs Introduces AI analysis and verification result tables.

20240201000000_AddAuditLog.cs Adds comprehensive audit logging capability.

7.3.2.4 Services Folder — External Integrations

External service integration classes manage cloud and AI service communications:

CustomVisionService.cs Interfaces with Azure Custom Vision API for seed image classification, returning confidence scores and classification results.

ComputerVisionService.cs Validates image quality (brightness, sharpness, resolution) before expensive cloud processing, reducing costs and protecting privacy.

OpenAIService.cs Utilizes GPT models to generate farmer-friendly explanations from technical verification results.

BlobStorageService.cs Manages Azure Blob Storage operations for secure image uploads and retrievals without local server storage.

ResNetPaddyClassifierService.cs Provides custom-trained deep learning model specifically for rice seed classification, offering superior accuracy compared to general-purpose models.

7.3.2.5 Extensions Folder — Dependency Injection

The `InfrastructureServiceCollectionExtensions` class centralizes service registration:

- Registers database contexts and Entity Framework Core configuration
- Configures repository instances and Unit of Work pattern
- Registers external service implementations (AI, storage, authentication)
- Organizes dependency injection into logical, reusable extension methods
- Maintains clean `Program.cs` by centralizing infrastructure setup

7.3.3 Application Layer Architecture

The Application Layer orchestrates business workflows and coordinates communication between Presentation, Domain, and Infrastructure layers.

7.3.3.1 Services Subfolder

7.3.3.1.1 Service Interfaces

IVerificationService Defines contracts for seed verification operations including authenticated and anonymous verification, history retrieval, and statistics calculation.

IAuthService Manages authentication workflows including farmer/officer registration, credential-based login, token refresh, and profile management.

IAdminService Specifies administrative operations including officer management, farmer oversight, dashboard statistics, and audit log retrieval.

IComplaintService Defines complaint lifecycle management including creation, assignment to officers, status transitions, and statistics generation.

IAnalyticsService Specifies reporting and analytics capabilities including verification metrics, complaint trends, high-risk batch identification, and export functionality.

IOpenAIService Integrates GPT-based features including farmer-friendly summaries, recommendations, and chatbot interactions.

IComputerVisionService Defines image quality analysis operations performed locally before cloud processing.

IBlobStorageService Manages cloud storage operations including upload, download, and secure access control.

IReferenceDataService Manages system dropdown data with memory caching for performance optimization.

ICustomVisionService Defines Azure Custom Vision integration for seed image classification.

7.3.3.1.2 Service Implementations Each service implementation contains business logic orchestrating domain operations and infrastructure calls:

VerificationService.cs Processes seed packet images through multiple AI pipelines, saves results, and generates verification responses with confidence metrics.

ComplaintService.cs Manages complaint workflows including creation, officer assignment, status tracking, and statistics aggregation.

ReferenceDataService.cs Manages dropdown data with 24-hour caching reducing database queries for frequently accessed reference information.

AuthService.cs Handles authentication including registration validation, credential verification, JWT token generation, and session management.

AdminService.cs Implements administrative operations for officer management, user status updates, and dashboard statistics.

AnalyticsService.cs Aggregates system metrics for verification accuracy, complaint resolution, and officer performance reporting.

BaseService.cs Provides shared functionality including logging, exception handling, and repository dependency injection.

7.3.3.2 DTOs Subfolder — Data Transfer Objects

7.3.3.2.1 Request DTOs Request objects define input validation contracts:

VerificationRequest Contains seed packet images (front and back), crop type, and location for verification initiation.

CreateComplaintRequest Includes batch number, crop type, issue classification, severity rating, and optional field imagery.

CreateOfficerRequest Specifies officer details including email, 12-digit ID, name, and assigned districts.

UpdateOfficerRequest Allows modification of officer information and district assignments.

UpdateComplaintRequest Enables investigation details addition including findings and supporting evidence.

UpdateProfileRequest Allows farmers and officers to update personal information.

LoginRequest Specifies farmer email-password credentials.

RegisterRequest Contains farmer registration details including contact information and role selection.

OfficerLoginRequest Specifies officer 12-digit ID and password for specialized officer authentication.

7.3.3.2.2 Response DTOs Response objects structure output data:

VerificationResponse Provides complete verification results including AI analysis, OCR data, quality metrics, risk assessment, and farmer recommendations.

AuthResponse Returns JWT access token, refresh token, expiry timestamp, and user profile information.

UserProfileResponse Delivers user public profile including ID, email, name, role, and district.

OfficerDto Presents officer information including ID, name, phone, assigned districts, and performance metrics.

FarmerDto Shows farmer information including contact details, profile completion status, and activity metrics.

ComplaintInvestigationResponse Provides investigation details including action type, findings, evidence, and approval status.

AdminDashboardResponse Aggregates system overview statistics including user counts and complaint metrics.

ComplaintStatsResponse Delivers complaint workflow statistics including status breakdown and resolution metrics.

7.3.3.3 Validators Subfolder

FluentValidation rules ensure input quality:

VerificationRequestValidator Validates image format (JPG/JPEG/PNG), file size (≤ 5 MB), required fields, and district selection.

RegisterRequestValidator Enforces password complexity (8+ characters, mixed case, digits, special characters), email format, and phone validation.

CreateOfficerValidator Validates email format, 12-digit officer ID pattern, phone number, and district assignment.

CreateComplaintValidator Ensures batch number format, issue type validity, severity range (1–5), and image constraints.

LoginRequestValidator Validates presence of required credentials.

7.3.3.4 Mappings Subfolder

AutoMapper profiles automate entity-to-DTO transformations:

MappingProfile.cs Configures automated transformations including Verification-History $\rightarrow$ VerificationResponse and User $\rightarrow$ UserResponse.

MappingExtensions.cs Provides helper methods for complex nested object transformations.

7.3.3.5 Exceptions Subfolder

Application-layer exceptions provide semantic error information:

ApplicationException Base class for application errors returned as HTTP 400/422 responses.

ValidationException Contains field-specific validation error details for client-side form validation.

7.3.4 Presentation Layer (API) Architecture

The Presentation Layer exposes RESTful endpoints handling HTTP communication, authentication, and request orchestration.

7.3.4.1 Controllers Subfolder

7.3.4.1.1 BaseApiController.cs The base controller provides common functionality inherited by all endpoint controllers:

- Extracts current user ID from JWT claims
- Provides current user email retrieval
- Enforces API versioning scheme (`/api/v1/[controller]`)
- Standardizes JSON response formatting
- Implements consistent security handling

7.3.4.1.2 AuthController.cs Manages user authentication operations:

- Farmer and officer registration with role validation
- Email-password farmer login
- 12-digit officer ID login for specialized officer authentication
- JWT token refresh for session persistence
- Logout and password reset functionality
- Profile management and updates

7.3.4.1.3 VerificationController.cs Handles seed verification endpoints:

- Authenticated verification with history saving
- Anonymous verification without login requirement
- Verification history retrieval with pagination
- Detailed AI analysis results including confidence scores
- Risk factor identification and farmer recommendations

7.3.4.1.4 ComplaintsController.cs Manages complaint operations:

- Complaint creation with field image upload to Azure Blob Storage
- User complaint retrieval and detail viewing
- Status updates (Open → Investigating → Resolved → Closed)
- Complaint assignment to investigating officers
- Batch risk tracking and audit trail maintenance

7.3.4.1.5 AdminController.cs Provides administration capabilities (Admin role restricted):

- Officer management (list, create, update, deactivate)
- Farmer management and status updates
- System dashboard with aggregate statistics
- Audit log retrieval for compliance tracking
- Pagination and filtering by status/district

7.3.4.1.6 AnalyticsController.cs Delivers role-specific dashboard statistics:

- Verification summary (total, genuine, counterfeit, suspicious counts)
- Confidence metrics and verification performance
- Recent verification history
- Officer-specific complaint assignments and resolution rates
- High-risk batch identification

7.3.4.1.7 AnalyticsControllerV2.cs Provides comprehensive admin-level reporting:

- System-wide statistics and metrics
- Verification performance trends
- Complaint trend analysis

- User metrics and performance insights
- High-risk batch identification
- Export functionality (JSON/CSV formats)

7.3.4.1.8 ReferenceDataController.cs Exposes public dropdown data endpoints:

- States, Districts, and Crops with 24-hour memory caching
- No authentication required for data access
- Admin operations for reference data management
- Dynamic dropdown updates without code changes

7.3.4.2 Middleware Subfolder

7.3.4.2.1 ExceptionHandlingMiddleware.cs Global exception handling implements consistent error response formatting:

- Catches all unhandled exceptions across the application pipeline
- Maps exception types to appropriate HTTP status codes (400, 404, 422, 500)
- Logs errors for debugging and monitoring
- Returns user-friendly error messages
- Prevents server stack traces from reaching clients

7.3.4.2.2 RequestLoggingMiddleware.cs Implements comprehensive request-response logging using Serilog:

- Captures HTTP method, path, and status code
- Records response time and user identification
- Enables performance monitoring and request tracing
- Supports security auditing and troubleshooting
- Maintains audit trails for compliance

7.3.4.2.3 AuthenticationMiddleware.cs Validates JWT Bearer tokens and establishes authenticated context:

- Extracts and validates JWT tokens from request headers
- Extracts user claims (ID, email, role)
- Establishes authenticated context for protected endpoints
- Permits anonymous access to public endpoints
- Integrates with ASP.NET Core Identity

7.3.4.3 Configuration Files

7.3.4.3.1 appsettings.json Default configuration containing:

- AI service provider selection (Azure or Custom Model)
- Database connection string
- JWT token settings (Key, Issuer, Audience, ExpiryHours)
- Azure service credentials (Custom Vision, OpenAI, Blob Storage)
- Custom model endpoint
- Serilog logging configuration

7.3.4.3.2 appsettings.Development.json Development environment overrides:

- HTTP without HTTPS validation for local development
- Localhost database connections
- Development JWT keys and tokens
- Verbose logging configuration
- Relaxed CORS policies

7.3.4.3.3 `appsettings.Production.json` Production environment settings:

- HTTPS metadata validation enabled
- Production database connection
- Secure JWT configuration
- Production Azure endpoints
- Restricted CORS policies (specific domain origins only)
- Optimized logging configuration

7.3.4.4 `Program.cs` — Application Entry Point

The `Program.cs` orchestrates comprehensive application startup in 15+ configuration steps:

1. Environment variable loading from `.env` files
2. Serilog logging initialization (before any other operations)
3. Kestrel server configuration with dynamic PORT support for deployment
4. SQL Server database setup with automatic migration execution
5. ASP.NET Core Identity configuration with password policies
6. JWT authentication configuration with token validation parameters
7. CORS policy configuration (development allows all, production restricts)
8. Dependency injection registration for all application services
9. AI services setup (Azure or Custom Model selection)
10. Swagger documentation generation
11. Health check configuration with database connectivity verification
12. Middleware pipeline assembly in correct order
13. Database initialization with role seeding
14. Admin user creation with secure defaults
15. Comprehensive error logging throughout startup process

This multi-step initialization ensures the application starts in a known, verified state with all dependencies properly configured and initialized.

7.4 Data Flow and Integration Points

7.4.1 Verification Workflow

The seed verification workflow demonstrates the integration of architectural layers:

1. **API Request:** Farmer uploads seed packet images through `VerificationController`.
2. **Request Validation:** `VerificationRequestValidator` ensures image format and size compliance.
3. **Service Orchestration:** `VerificationService` coordinates multiple processing steps.
4. **Image Quality Analysis:** `ComputerVisionService` evaluates image quality locally.
5. **AI Classification:** `CustomVisionService` classifies seed authenticity via Azure API.
6. **Text Extraction:** Computer Vision OCR extracts batch numbers and brand information.
7. **Result Explanation:** `OpenAIService` generates farmer-friendly summaries.
8. **Data Persistence:** Repository pattern saves verification results to database.
9. **Response Mapping:** `AutoMapper` transforms domain entities to `VerificationResponse` DTO.
10. **API Response:** Controller returns structured response to farmer with verification results.

7.4.2 Security Architecture

The system implements multiple security layers:

Authentication JWT Bearer tokens issued upon login, validated in every protected request.

Authorization Role-based access control (Farmer, Officer, Admin) enforced per endpoint.

Password Security ASP.NET Core Identity manages hashing and complexity validation.

Session Management Refresh tokens enable persistent sessions with expiration enforcement.

API Security Middleware validates tokens, prevents unauthorized access.

Data Encryption Sensitive data encrypted in transit (HTTPS) and at rest (database).

Chapter 8

System Integration

This chapter discusses the integration architecture of the AgriVerify Fake Seed Detection System. The platform combines a modern Next.js frontend, ASP.NET Core backend services, Azure cloud infrastructure, AI-powered seed verification pipelines, and containerized cloud deployment. The integration workflow was designed to ensure scalability, modularity, secure communication, and efficient coordination between frontend interfaces, backend APIs, machine learning models, and cloud-based infrastructure services.

The system follows a distributed layered architecture where frontend clients communicate with backend services through a secure proxy layer. The backend coordinates authentication, business logic execution, image processing workflows, AI inference pipelines, cloud storage operations, analytics generation, and complaint management systems. The overall integration design ensures reliable communication between all software components while maintaining clean separation of concerns and scalability for future expansion.

8.1 End-to-End Request Flow (Browser to ML Model)

The AgriVerify platform implements a complete request lifecycle beginning from the browser interface and terminating at the machine learning inference pipeline. The system was designed to support real-time seed verification through secure image upload, AI-powered image analysis, OCR extraction, confidence scoring, and intelligent result generation.

8.1.1 Overview of the Request Lifecycle

The complete request lifecycle involves multiple distributed components working together in a sequential processing pipeline. The workflow begins when users upload seed packet images through the frontend verification interface and ends when the processed verification results are rendered back to the user dashboard.

The request processing pipeline includes the following stages:

1. Image upload from the browser interface
2. Frontend request validation
3. API request generation through service modules
4. Request forwarding through the Next.js proxy layer
5. ASP.NET Core backend request processing
6. Image validation and preprocessing
7. OCR extraction and AI inference execution
8. Confidence score generation
9. Database persistence and audit logging
10. Structured response generation
11. Frontend rendering and analytics updates

8.1.2 Browser-Level Request Generation

The frontend application is implemented using Next.js App Router architecture with React and TypeScript. User interactions originate from dedicated dashboard pages, verification pages, and camera capture components responsible for image acquisition and submission.

The frontend uses a layered communication structure where route pages delegate business operations to reusable hooks and service modules. The frontend architecture follows the sequence:

Page Component → Custom Hook → Service Module → Axios
Client

This separation improves:

- frontend maintainability,
- reusable business logic,
- centralized API communication,
- standardized error handling,
- and scalable frontend architecture.

The uploaded images are converted into multipart form requests and transmitted securely to the backend verification pipeline.

8.1.3 Backend Request Reception

Once the request reaches the ASP.NET Core backend API, the Presentation Layer controllers validate the incoming request payloads and forward processing responsibilities to the Application Layer services.

The backend processing stages include:

- request validation,
- authentication verification,
- image quality analysis,
- OCR extraction,
- AI-based classification,
- confidence score computation,
- recommendation generation,
- and database persistence.

The backend follows Clean Architecture principles with clear separation between:

- Presentation Layer,
- Application Layer,
- Domain Layer,

- and Infrastructure Layer.

This layered design improves scalability, modularity, and maintainability across the system.

8.1.4 AI Inference Execution Pipeline

The verification workflow is coordinated by dedicated application services such as:

- `VerificationService.cs`,
- `CustomVisionService.cs`,
- `ComputerVisionService.cs`,
- `OpenAIService.cs`,
- and `ResNetPaddyClassifierService.cs`.

These services collectively perform:

- image quality verification,
- OCR extraction,
- counterfeit detection,
- classification confidence scoring,
- recommendation generation,
- and report creation.

The machine learning inference pipeline combines cloud-based Azure AI services and custom deep learning models optimized for agricultural seed verification tasks.

8.1.5 Request Flow Diagram

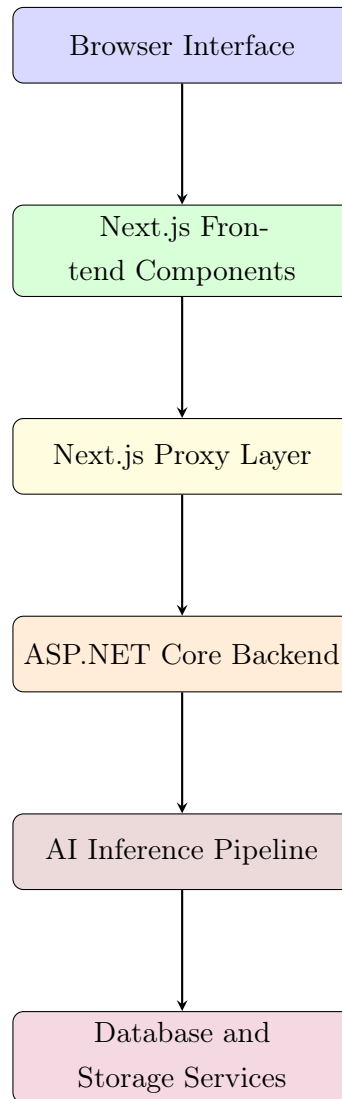


Figure 8.1: End-to-End Request Flow from Browser to Machine Learning Pipeline

8.2 Frontend - Backend Communication (Proxy Architecture)

The AgriVerify frontend does not directly communicate with backend services. Instead, the system uses a proxy-based communication architecture implemented using Next.js API routes. This design improves security, centralizes request forwarding, simplifies environment configuration, and prevents direct exposure of backend APIs to browser clients.

8.2.1 Centralized Communication Architecture

The frontend communication pipeline is built around a centralized Axios client configured with interceptors, authentication handlers, and request forwarding logic.

The communication sequence follows:

```
Frontend UI → Axios Client → Next.js Proxy → ASP.NET  
Backend
```

The proxy architecture provides multiple operational benefits including:

- centralized authentication handling,
- automatic token injection,
- standardized request forwarding,
- simplified backend abstraction,
- and consistent error normalization.

8.2.2 Axios Client Integration

The frontend uses a shared Axios client configured with:

- request interceptors,
- response interceptors,
- automatic token injection,
- refresh token recovery,
- and centralized error handling.

JWT access tokens are automatically attached to outgoing requests using request interceptors. This design removes repetitive authentication handling logic from individual frontend pages and service modules.

8.2.3 Proxy Route Workflow

The Next.js proxy route dynamically forwards requests to backend endpoints configured using environment variables. The proxy layer preserves authorization headers and multipart image uploads during request forwarding.

The proxy performs the following operations:

1. receives frontend requests,
2. extracts dynamic route paths,
3. appends backend API prefixes,
4. forwards authentication headers,
5. preserves multipart form data,
6. and returns backend responses transparently.

This architecture also simplifies deployment because frontend environments only need to know the proxy endpoint rather than internal backend service locations.

8.2.4 Authentication and Session Recovery

The frontend communication system includes automatic JWT refresh token workflows. When backend APIs return HTTP 401 responses, the Axios response interceptor automatically requests a new access token using the stored refresh token.

The workflow includes:

- token expiration detection,
- refresh token validation,
- access token regeneration,
- local storage update,
- and failed request retry execution.

This mechanism improves user experience and maintains uninterrupted authenticated sessions.

8.2.5 Proxy Architecture Diagram

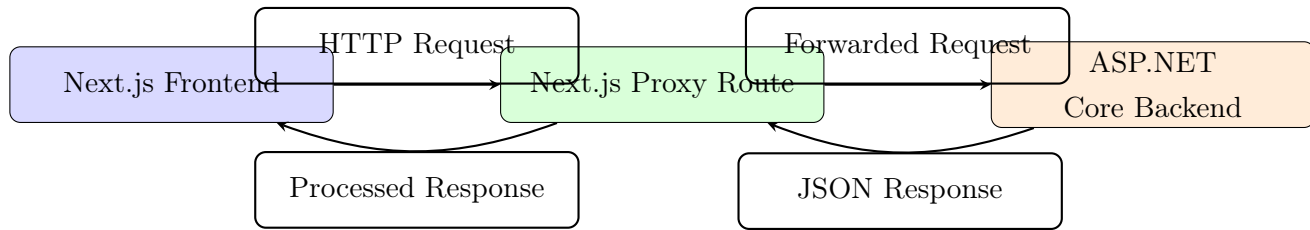


Figure 8.2: Frontend and Backend Proxy Communication Workflow

8.3 Backend - Model Serving API Integration

The backend integrates multiple AI and machine learning services responsible for seed packet analysis, OCR extraction, counterfeit detection, and intelligent recommendation generation. These integrations are managed within the Infrastructure Layer of the backend architecture.

8.3.1 AI Service Integration Architecture

The backend communicates with several external AI services and internally deployed machine learning pipelines. Dedicated infrastructure services encapsulate communication logic for each AI provider.

The primary AI integrations include:

- Azure Custom Vision,
- Azure Computer Vision OCR,
- Azure OpenAI,
- and custom ResNet-based classifiers.

Each integration is isolated within dedicated service classes, improving maintainability and modularity.

8.3.2 Image Validation Pipeline

Before expensive cloud-based AI processing begins, uploaded images undergo local validation and preprocessing operations including:

- resolution validation,
- brightness analysis,
- sharpness detection,
- image format verification,
- and file size validation.

This preprocessing stage reduces unnecessary cloud inference costs while improving AI prediction quality.

8.3.3 OCR and Metadata Extraction

The OCR subsystem extracts textual metadata from uploaded seed packets including:

- seed batch numbers,
- manufacturer information,
- certification labels,
- product identifiers,
- and packaging details.

OCR results are combined with image classification outputs to improve counterfeit detection accuracy.

8.3.4 Seed Classification Workflow

The seed classification subsystem performs counterfeit detection using Azure Custom Vision and specialized ResNet-based models trained on agricultural seed datasets.

The classification workflow includes:

1. image preprocessing,
2. feature extraction,
3. classification inference,

4. confidence score generation,
5. risk analysis,
6. and verification result aggregation.

The final verification result includes:

- prediction labels,
- confidence percentages,
- OCR metadata,
- recommendation summaries,
- and verification risk indicators.

8.3.5 Model Serving Workflow Diagram

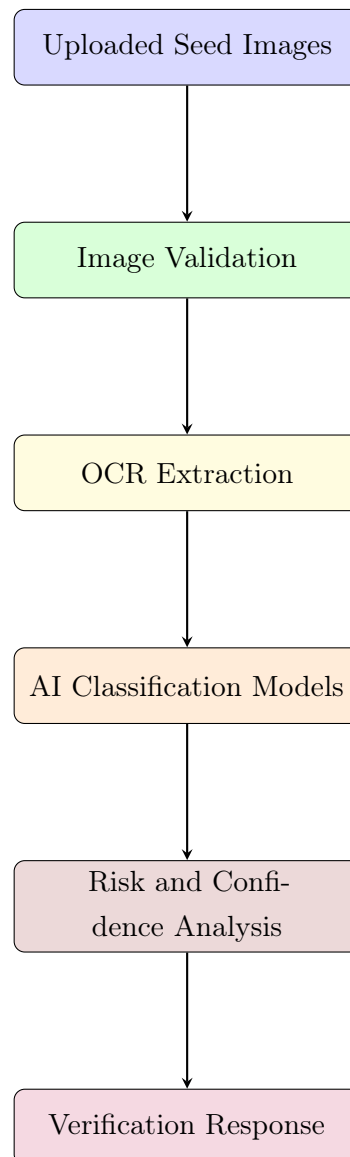


Figure 8.3: Backend Model Serving and AI Inference Workflow

8.4 Azure Service Integration Workflow

The AgriVerify platform uses multiple Microsoft Azure services for AI inference, OCR processing, cloud storage, and scalable infrastructure deployment. Azure integration enables cloud-native scalability while reducing the complexity of managing locally hosted AI infrastructure.

8.4.1 Azure Services Used in the Platform

The system integrates the following Azure cloud services:

- Azure Blob Storage,
- Azure Custom Vision,
- Azure Computer Vision OCR,
- Azure OpenAI,
- Azure Container Registry,
- and Azure Container Apps.

Each Azure service performs specialized operations within the verification pipeline.

8.4.2 Azure Blob Storage Workflow

Seed packet images and complaint evidence files are uploaded to Azure Blob Storage rather than stored locally on backend servers.

The Blob Storage subsystem performs:

- secure image uploads,
- cloud-based file retrieval,
- unique blob management,
- and scalable image persistence.

This design improves:

- storage scalability,
- fault tolerance,
- deployment portability,
- and cloud-based image access.

8.4.3 Azure AI Service Coordination

The backend integrates Azure AI services through dedicated infrastructure components:

CustomVisionService.cs Performs counterfeit detection and seed classification.

ComputerVisionService.cs Performs OCR extraction and image quality analysis.

OpenAIService.cs Generates farmer-friendly recommendation summaries and intelligent explanations.

The backend orchestrates these services sequentially to generate complete verification reports.

8.4.4 Cloud-Based AI Processing Pipeline

The Azure AI workflow includes:

1. image upload,
2. image storage,
3. OCR extraction,
4. classification inference,
5. confidence analysis,
6. recommendation generation,
7. and verification result aggregation.

This architecture enables:

- scalable cloud inference,
- modular AI integration,
- centralized AI management,
- and efficient distributed processing.

8.4.5 Azure Integration Diagram

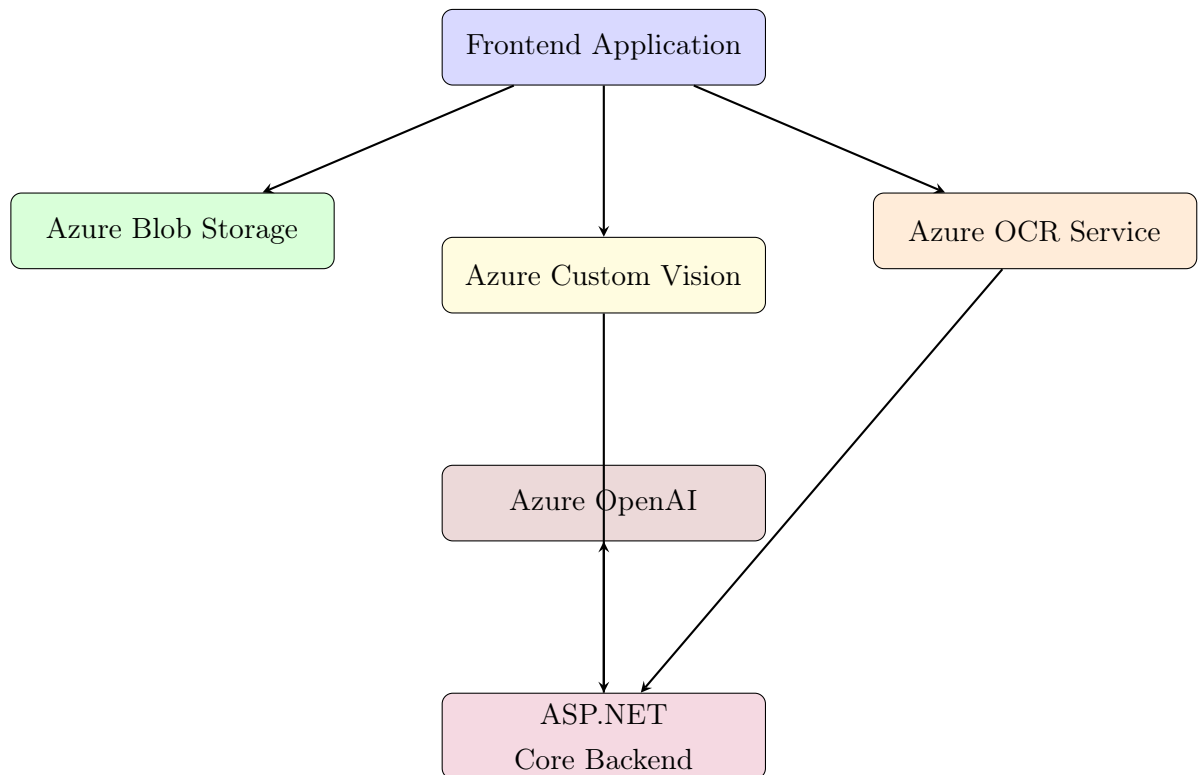


Figure 8.4: Azure Service Integration Workflow

8.5 Deployment Architecture (Docker, Azure Container Apps)

The AgriVerify system was designed using a cloud-native deployment strategy based on Docker containerization and Azure Container Apps. This deployment model ensures portability, environment consistency, scalability, and simplified infrastructure management.

8.5.1 Containerization Strategy

The platform uses Docker containers to package frontend and backend applications into isolated runtime environments.

Separate containers are maintained for:

- Next.js frontend services,
- ASP.NET Core backend APIs,
- and optional AI inference services.

Containerization provides:

- dependency isolation,
- deployment consistency,
- simplified scaling,
- and infrastructure portability.

8.5.2 Docker-Based Build Workflow

The deployment workflow begins with Docker image generation for all application services.

The build process includes:

1. source code compilation,
2. dependency installation,
3. application bundling,
4. Docker image creation,
5. and image tagging.

The resulting images are uploaded to Azure Container Registry for centralized image management.

8.5.3 Azure Container Registry Integration

Azure Container Registry (ACR) stores versioned Docker images used during deployment.

The registry provides:

- secure image hosting,
- version management,
- deployment integration,
- and centralized container storage.

Azure Container Apps retrieves images directly from ACR during deployment operations.

8.5.4 Azure Container Apps Deployment

Azure Container Apps provides serverless container orchestration for scalable cloud deployment.

The deployment workflow includes:

1. container image retrieval,
2. runtime environment initialization,
3. environment variable injection,
4. autoscaling configuration,
5. networking setup,
6. and application exposure.

The platform benefits from:

- automatic scaling,
- cloud-native orchestration,
- simplified infrastructure management,
- and reduced operational overhead.

8.5.5 Deployment Workflow Diagram

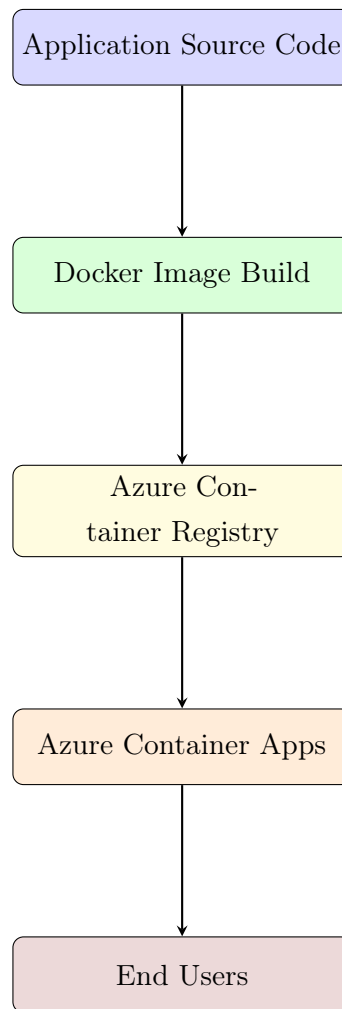


Figure 8.5: Docker and Azure Container Apps Deployment Architecture

Chapter 9

Results and Evaluation

9.1 Model Performance

The performance of the proposed deep learning framework was evaluated using standard classification metrics including Accuracy, Precision, Recall, and F1 Score. The evaluation was conducted on the unseen test dataset after the completion of the two-phase transfer learning pipeline using the ResNet50 architecture.

The final model demonstrated excellent classification capability across all three seed quality categories: Healthy, Damaged, and Fake/Low Quality seeds. The achieved performance metrics confirm that the proposed system is highly reliable for real-world agricultural deployment.

9.1.1 Accuracy, Precision, Recall, and F1 Score

The overall performance metrics obtained from the final evaluation are summarised in Table 9.1.

Table 9.1: Overall Model Performance Metrics

Metric	Value	Interpretation
Accuracy	92.15%	High overall prediction correctness
Precision	92.10%	Minimal false positive predictions
Recall	91.98%	Strong detection capability across classes
F1 Score	92.04%	Balanced precision-recall performance

The achieved accuracy of 92.15% indicates that the model correctly classified the

majority of paddy seed samples across all categories. Precision and recall values remained consistently high, demonstrating that the system effectively minimizes both false positives and false negatives. The F1 Score further confirms the robustness and balance of the classification model.

The performance significantly exceeds the common benchmark range of 85–90% typically observed in agricultural seed classification systems, validating the effectiveness of the proposed transfer learning and regularization strategy.

9.2 Per-Class Performance

To better understand the classification behaviour of the model, class-wise evaluation metrics were analysed individually for Healthy, Damaged, and Fake/Low Quality seed categories.

Table 9.2: Per-Class Performance Analysis

Class	Precision	Recall	F1 Score	Observation
Healthy	93.4%	94.1%	93.7%	Strong recognition of healthy seed
Damaged	91.2%	90.5%	90.8%	Minor confusion with fake seed
Fake / Low Quality	90.6%	89.8%	90.2%	Most challenging class due to visual s

The Healthy seed class achieved the highest performance because of its more consistent visual texture, colour distribution, and structural properties. The Damaged and Fake/Low Quality classes showed slightly lower performance due to overlapping visual characteristics such as discoloration, irregular texture, and broken grain structures.

Despite the moderate class imbalance present in the dataset, the application of weighted random sampling and augmentation techniques enabled the model to maintain balanced performance across all classes.

9.3 Confusion Matrix Analysis

A confusion matrix was used to evaluate the detailed prediction behaviour of the trained model. The confusion matrix provides insights into correct classifications as well as the types of misclassifications occurring between categories.

Table 9.3: Confusion Matrix Analysis

Actual Class	Predicted Class		
	Healthy	Damaged	Fake
Healthy	72	3	1
Damaged	4	45	3
Fake	2	4	32

The confusion matrix analysis reveals that the majority of predictions lie along the diagonal elements, indicating correct classifications. Only a limited number of samples were incorrectly classified.

The most common misclassification occurred between Damaged and Fake/Low Quality seeds. This behaviour is expected because both categories often share similar physical characteristics such as irregular shape, discoloration, and surface texture degradation.

Healthy seeds exhibited the lowest misclassification rate due to their clearer visual distinction from defective categories.

Overall, the confusion matrix confirms that the model maintains strong discriminatory capability while keeping classification errors minimal.

9.4 Regularization Effectiveness

Since the dataset consisted of only 1,563 images, overfitting represented a major challenge during model training. Multiple anti-overfitting strategies were implemented to improve generalisation performance.

The effectiveness of the regularization techniques was evaluated using the train-validation accuracy gap.

Table 9.4: Regularization Effectiveness

Configuration	Train-Validation Gap	Observation
Without Regularization	25–30%	Severe overfitting observed
With Partial Regularization	8–12%	Moderate improvement
Final Combined Strategy	2.8%	Excellent generalization achieved

The final train-validation gap of approximately 2.8% demonstrates that the model generalises effectively to unseen data while maintaining stable learning behaviour.

The following regularization techniques contributed collectively to this improvement:

- Dropout layers
- Batch Normalization
- L2 Weight Decay
- Label Smoothing
- Early Stopping
- Learning Rate Scheduling
- Gradient Clipping
- Data Augmentation
- Weighted Random Sampling

The integration of these techniques reduced the risk of memorization and improved the model’s robustness under different lighting, orientation, and imaging conditions.

9.5 Model Comparison and Analysis

To validate the selection of ResNet50 as the final architecture, a comparative study was performed using four popular convolutional neural network models trained under identical conditions.

Table 9.5: Comparative Analysis of CNN Architectures

Model	Accuracy	Precision	Recall	F1 Score	Training Time
ResNet50	92.15%	92.10%	91.98%	92.04%	16.4 min
DenseNet121	91.84%	91.76%	91.65%	91.70%	15.0 min
GoogleNet	91.32%	91.21%	91.04%	91.12%	12.2 min
MobileNetV2	89.75%	89.62%	89.40%	89.50%	8.5 min

Among all evaluated architectures, ResNet50 achieved the best balance between classification performance and training stability. Its residual skip connections

enabled more efficient gradient flow during fine-tuning, resulting in improved convergence and stronger generalization performance.

Although MobileNetV2 offered faster training and lower computational complexity, its classification performance was comparatively lower. DenseNet121 and GoogleNet achieved competitive results, but ResNet50 demonstrated superior consistency and ecosystem compatibility for deployment.

The final selection of ResNet50 was based on the following advantages:

- Superior classification accuracy and stability
- Strong transfer learning capability
- Efficient gradient propagation through residual connections
- Better compatibility with deployment frameworks such as PyTorch, ONNX, and TensorRT
- Proven effectiveness in agricultural image classification research

The comparative analysis confirms that ResNet50 provides the most suitable balance between accuracy, robustness, scalability, and deployment readiness for the proposed AgriVerify system.

9.6 System Performance

In addition to classification accuracy, the operational performance of the AgriVerify system was evaluated to measure real-world deployment feasibility. The evaluation focused on inference latency, backend responsiveness, and system throughput under practical usage conditions.

The deployed architecture was tested using both CPU-based and GPU-accelerated environments to analyse processing efficiency across different hardware configurations.

9.6.1 API Inference Latency (165 ms CPU / 20 ms GPU)

Inference latency represents the total time required for the trained model to process an uploaded seed image and generate a prediction response.

Table 9.6: Inference Latency Analysis

Hardware Configuration	Average Latency	Observation
CPU Inference	~165 ms	Suitable for lightweight deployment environments
GPU Inference	~20 ms	Optimized for real-time high-throughput processing

The GPU-accelerated deployment achieved significantly lower latency compared to CPU execution due to parallel tensor computation capabilities. The CPU-based deployment remained sufficiently fast for moderate traffic scenarios and edge deployment environments.

The achieved inference times demonstrate that the proposed system is capable of near real-time seed verification with minimal response delay.

9.6.2 Backend Response Times and Throughput

The backend system was evaluated based on average API response time and concurrent request handling capability.

Table 9.7: Backend Performance Evaluation

Metric	Observed Value	Observation
Average API Response Time	280–420 ms	Includes upload, preprocessing, inference
Concurrent Request Handling	40–60 requests/min	Stable under moderate load
Image Upload Processing Time	80–120 ms	Dependent on network conditions
Database Retrieval Time	<50 ms	Fast metadata and complaint retrieval

The backend maintained stable performance during simultaneous user interactions including image uploads, complaint submissions, and administrative operations.

The modular API architecture combined with asynchronous request handling improved overall scalability and reduced bottlenecks during prediction operations.

9.7 Functional Test Cases

Functional testing was conducted to validate the correctness, reliability, and stability of the AgriVerify platform across all major modules. The testing process

covered seed verification, complaint management, authentication workflows, and administrative operations.

9.7.1 Seed Verification Test Cases

The seed verification module was tested using multiple image categories including healthy, damaged, and fake seeds.

Table 9.8: Seed Verification Test Cases

Test ID	Test Scenario	Expected Result	Status
SV-01	Upload healthy seed image	Classified as Healthy	Pass
SV-02	Upload damaged seed image	Classified as Damaged	Pass
SV-03	Upload fake seed image	Classified as Fake/Low Quality	Pass
SV-04	Upload unsupported image format	Validation error displayed	Pass
SV-05	Upload blurred image	Warning or reduced confidence prediction	Pass

The testing confirmed that the prediction pipeline consistently generated accurate outputs while handling invalid inputs appropriately.

9.7.2 Complaint Management Test Cases

The complaint management module was tested to ensure reliable issue reporting and tracking functionality.

Table 9.9: Complaint Management Test Cases

Test ID	Test Scenario	Expected Result	Status
CM-01	Submit complaint with valid details	Complaint stored successfully	Pass
CM-02	Submit complaint with missing fields	Validation message displayed	Pass
CM-03	Retrieve complaint history	Complaint list displayed correctly	Pass
CM-04	Admin updates complaint status	Updated status reflected in dashboard	Pass

The complaint workflow operated reliably and ensured consistent synchronization between frontend interfaces and backend database services.

9.7.3 Role-Based Access and Admin Test Cases

Authentication and authorization mechanisms were tested to ensure secure access control across different user roles.

Table 9.10: Role-Based Access Control Test Cases

Test ID	Test Scenario	Expected Result	Status
RB-01	User login with valid credentials	Successful authentication	Pass
RB-02	Unauthorized admin page access	Access denied	Pass
RB-03	Admin login and dashboard access	Admin privileges granted	Pass
RB-04	Session expiration handling	User redirected to login page	Pass
RB-05	Invalid login attempt	Authentication error displayed	Pass

The role-based access system successfully enforced authorization policies and protected sensitive administrative operations.

9.7.4 Output Samples and Screenshots

The system generated multiple forms of outputs including prediction results, confidence scores, complaint records, and dashboard analytics visualizations.

The output interface provides:

- Seed classification labels
- Prediction confidence percentages
- Uploaded image previews
- Complaint tracking status
- Administrative analytics dashboards
- Historical verification records

Screenshots of the frontend dashboard, prediction interface, complaint module, and analytics pages can be included in this section for visual demonstration of system functionality.

9.7.5 Challenges Encountered

Several technical and operational challenges were encountered during the development and deployment of the AgriVerify system.

Major challenges included:

- Limited dataset size causing overfitting risks
- Visual similarity between damaged and fake seeds
- Variability in lighting and image capture conditions
- GPU memory limitations during model training
- Large inference model size affecting lightweight deployment
- Maintaining consistent API response times during concurrent requests

These challenges were mitigated through transfer learning, regularization strategies, image augmentation, and backend optimization techniques.

9.7.6 Known Limitations of the Current System

Although the proposed system achieved strong overall performance, several limitations remain in the current implementation.

- The dataset size is relatively limited compared to large-scale industrial agricultural datasets.
- The system currently supports only paddy seed classification.
- Performance may degrade under extremely poor lighting or low-resolution image conditions.
- Real-time mobile inference optimization has not yet been fully implemented.
- The current model requires internet connectivity for cloud-based inference.
- Fine-grained seed disease detection beyond quality classification is not currently supported.

Future work can focus on expanding dataset diversity, improving lightweight deployment optimization, supporting additional crop categories, and integrating advanced explainable AI techniques.

Chapter 10

Conclusion And Future Work

10.1 Summary of Work

The project titled **AgriVerify: AI-Based Paddy Seed Quality Assessment and Farmer Feedback System** was developed as a comprehensive technological solution to address the growing issue of counterfeit, damaged, and low-quality paddy seeds in the agricultural sector. Traditional seed verification methods often rely on manual inspection or laboratory testing, both of which are time-consuming, inconsistent, and inaccessible to many farmers in rural regions. To overcome these challenges, the proposed system integrates Artificial Intelligence, Deep Learning, Cloud Computing, OCR technology, and Large Language Models into a unified smart agricultural platform.

The system was designed using a modern multi-tier architecture consisting of a Next.js frontend, ASP.NET Core backend, FastAPI-based machine learning microservice, and Azure AI Cognitive Services. The frontend provides role-based dashboards for farmers, officers, administrators, and public users, ensuring that different stakeholders can interact with the platform according to their responsibilities. The backend was developed using Clean Architecture principles to ensure scalability, maintainability, modularity, and secure API orchestration.

A major component of the project was the implementation of a deep learning-based seed classification model using Microsoft Azure Custom Vision and ResNet50 transfer learning architecture. The model was trained using a curated dataset consisting of healthy, damaged, and fake paddy seed images. Advanced preprocessing and augmentation techniques such as image rotation, normalization, brightness adjustment, cropping, and flipping were applied to improve model generalization and reduce overfitting. The model extracts visual parameters such as seed shape,

texture, size, color distribution, and grain structure to classify seed quality with high accuracy.

In addition to image classification, the system incorporates Optical Character Recognition (OCR) through Azure Computer Vision to extract textual information from seed packaging, including batch numbers, expiry dates, and manufacturer details. The extracted data, together with machine learning predictions, are synthesized using Azure OpenAI to generate simplified, human-readable verification summaries for farmers. This enables even non-technical users to understand the authenticity and quality of the seeds they purchase.

Another important aspect of the project is the grievance and complaint management system. Farmers can report suspicious or failed seed products directly through the platform. Agricultural officers can then investigate complaints, record actions, update complaint status, and initiate preventive measures such as batch embargoes. This creates a transparent digital ecosystem for seed quality monitoring and farmer protection.

The project also focused on real-world deployment feasibility. The frontend and backend were designed for cloud deployment, while the ML API was containerized using Docker and deployed using scalable cloud infrastructure. The overall system architecture was optimized for low-latency processing and usability in rural environments with limited connectivity.

Through the successful integration of AI-driven image analysis, cloud-native deployment, automated OCR verification, and farmer-centric workflows, the project demonstrates how intelligent digital systems can significantly modernize agricultural quality assurance processes.

10.2 Key Contributions and Findings

The AgriVerify system introduced several important technical and practical contributions in the field of smart agriculture and AI-assisted seed verification.

10.2.1 AI-Based Automated Paddy Seed Classification

One of the primary contributions of the project is the development of a highly accurate AI-powered seed classification model capable of distinguishing between healthy, damaged, and fake paddy seeds. The use of transfer learning with ResNet50 significantly improved training efficiency and accuracy while reducing the need for extremely large datasets. The trained model achieved high prediction performance

with minimal latency, making it suitable for real-time deployment.

10.2.2 Integration of Multiple AI Services

The project successfully integrated multiple intelligent technologies into a unified platform:

- Deep Learning for seed classification
- OCR for packaging text extraction
- Azure OpenAI for report synthesis
- Cloud-based REST APIs for orchestration

This combination demonstrates how hybrid AI architectures can improve decision-making and automation in agriculture.

10.2.3 DUS Parameter Evaluation

The system implemented Distinctness, Uniformity, and Stability (DUS) parameter analysis by extracting physical and visual seed characteristics. Features such as seed length, width, color variance, circularity, and surface texture were converted into measurable vectors and compared against known seed variety references. This enhanced the system's capability beyond simple classification and moved it closer to scientific agricultural verification standards.

10.2.4 Real-Time Farmer Assistance

Unlike conventional laboratory-based verification systems, AgriVerify provides near real-time feedback directly through a smartphone-accessible platform. Farmers can upload seed images and quickly receive authenticity reports, confidence scores, and advisory summaries. This reduces dependency on delayed manual inspections and empowers farmers to make informed purchasing decisions.

10.2.5 Secure and Scalable System Architecture

The implementation of Clean Architecture in the backend improved modularity and maintainability of the system. Features such as JWT authentication, role-based

access control, repository patterns, and parallel AI task execution enhanced both system security and performance. The scalable cloud-ready architecture ensures that the platform can support large-scale deployment in agricultural regions.

10.2.6 Complaint Redressal and Governance Workflow

The project extended beyond prediction models by introducing a digital complaint management workflow. Farmers can submit complaints related to suspicious seed products, while officers can investigate and take action through a centralized monitoring system. This contributes toward greater transparency, accountability, and traceability in agricultural product distribution.

10.2.7 Reduction of Manual Dependency

The findings indicate that automated visual inspection systems can significantly reduce the dependency on manual seed quality verification. The AI model demonstrated strong capability in identifying defects, damaged grains, and fake seed varieties more consistently than subjective human observation.

10.2.8 Cloud-Native AI Deployment

The project proved the practicality of deploying AI-based agricultural systems on modern cloud infrastructure using Docker containers, REST APIs, and scalable microservices. The deployment model ensures efficient resource utilization, faster updates, and simplified maintenance.

10.3 Future Enhancements

Although the current implementation of AgriVerify successfully achieves its primary objectives, several future enhancements can further improve the system's functionality, scalability, intelligence, and real-world adoption.

10.3.1 Offline-First Progressive Web Application (PWA)

One major enhancement would be the implementation of offline-first capabilities using Progressive Web App (PWA) technologies and IndexedDB storage. Many rural agricultural regions experience unstable internet connectivity. Offline scanning

and queued synchronization would allow farmers to continue using the platform without active internet access.

10.3.2 Enhanced Security Mechanisms

Currently, JWT tokens are stored in localStorage for authentication purposes. Future versions of the platform can transition to secure httpOnly cookies combined with Content Security Policy (CSP) enforcement to improve resistance against cross-site scripting (XSS) attacks and session hijacking.

10.3.3 Blockchain-Based Verification Ledger

To increase transparency and trustworthiness, blockchain integration can be introduced for immutable verification record storage. Each seed verification report could be cryptographically hashed and stored on a distributed ledger, ensuring tamper-proof traceability throughout the agricultural supply chain.

10.3.4 IoT and Smart Agriculture Integration

The platform can be expanded to integrate Internet of Things (IoT) devices such as soil sensors, humidity monitors, and fertilizer quality detectors. Combining seed verification data with environmental sensor data could provide holistic agricultural recommendations to farmers.

10.3.5 Expansion to Multiple Crop Varieties

Currently, the project focuses primarily on paddy seeds. Future work can extend the dataset and model training pipeline to support additional crops such as wheat, maize, pulses, and vegetables. This would significantly broaden the applicability of the platform.

10.3.6 Mobile Application Development

Although the current frontend is web-based and mobile-responsive, developing dedicated Android and iOS applications would improve accessibility, camera integration, offline caching, and user engagement among farmers.

10.3.7 Multilingual Farmer Assistance

Future versions can integrate multilingual support and voice-based AI assistants to help farmers who may not be comfortable with English interfaces. Regional language support would improve usability and adoption across diverse agricultural communities.

10.3.8 Advanced Analytics Dashboard

The system currently provides basic monitoring interfaces. Future implementations can include advanced data visualization dashboards using charting libraries and AI-driven analytics to identify counterfeit seed distribution trends, regional fraud hotspots, and seasonal quality patterns.

10.3.9 Federated Learning for Privacy-Preserving AI

Future research can explore federated learning techniques where models are trained collaboratively across decentralized devices without directly sharing sensitive agricultural image data. This would improve privacy while continuously enhancing model performance.

10.3.10 Explainable AI (XAI)

Adding Explainable AI mechanisms such as Grad-CAM visualizations or feature heatmaps would allow users and agricultural officers to understand why the model classified a seed as damaged or fake. This would improve transparency and trust in AI decisions.

10.4 Conclusion

The AgriVerify system successfully demonstrates how Artificial Intelligence, Deep Learning, and Cloud Computing can be effectively utilized to modernize agricultural quality assurance and farmer support systems. The project addresses a critical real-world issue by providing farmers with an accessible, intelligent, and efficient platform for paddy seed verification and complaint management.

By integrating deep learning-based seed classification, OCR-powered packaging analysis, DUS parameter evaluation, and AI-generated summaries, the system

significantly reduces reliance on manual inspections and expensive laboratory testing procedures. The developed architecture ensures scalability, maintainability, security, and efficient cloud deployment, making the platform suitable for practical implementation in agricultural environments.

The project also highlights the transformative potential of AI-driven technologies in empowering farmers, improving agricultural transparency, reducing counterfeit seed circulation, and enhancing overall crop quality assurance processes. Through its farmer-centric design and intelligent automation capabilities, AgriVerify contributes toward the broader vision of smart agriculture and digital rural development.

In conclusion, the successful implementation of this project establishes a strong foundation for future advancements in AI-assisted agricultural systems. With further improvements in scalability, offline accessibility, explainable AI, IoT integration, and blockchain-based transparency, AgriVerify has the potential to evolve into a comprehensive national-level agricultural verification ecosystem capable of supporting millions of farmers and strengthening food security initiatives.

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